05 — Use Cases and Process Flows

Prepared 9 November 2026 for teacher review on 10 November. **Draft; not approved or implemented.**

Correct count: UC01–UC15 and F01–F18. F18 covers optional participation/history/group policies; it must not disappear from the catalogue just because it is optional.

The compact v2 flows are summaries. The detailed v1 Mermaid below is inherited reference material, not confirmation that every latest chat/image correction was incorporated. F12 includes reauthorization and waiting/recovered outcomes, but its media-subscription failure and closing-state guards still need explicit review before design approval.

Source: 08-use-cases.md

Use-case catalogue

Version: 0.1 · **Prepared:** 9 November 2026 · **Status:** Proposed review draft; no teacher approval or product test execution claimed.

1. System boundary and catalogue

A use case is an actor goal, not a function count or mandatory runtime sequence. Preserve these UC IDs across the board, stories and tests.

Use case	Primary actor	Goal / activities
UC01	Teacher	Create class and publish joining information — F04
UC02	Student, Teacher for approval	Request and obtain admission — F05
UC03	Teacher, admitted Student	Publish/receive media and playback controls — F06/F07
UC04	Teacher	Manage roster and participant access — F08
UC05	SYS/NET with policy permission	Create/maintain profiles and exceptions — F09
UC06	SYS/NET or authorized class automation	Apply and verify supported restrictions — F10/F11
UC07	Student	Recover class connection — F12
UC08	Teacher and backend automation	End class, revoke access and restore policy — F13
UC09	SYS/NET or expiry automation	Emergency/expiry recovery — F14
UC10	Authorized operator	Diagnose/recover local services — F15
UC11	Teacher, admitted Student	Share/access permitted materials — F16
UC12	SYS/NET with policy permission	Schedule/import/export/local lists — F17
UC13	Teacher, Student, scoped policy administrator	Optional participation, history or group policy — F18

Use case	Primary actor	Goal / activities
UC14	Authorized campus installer/NET	Prepare and verify campus deployment — F02
UC15	Authorized portable installer/NET	Prepare and verify portable deployment — F03

2. Use-case detail convention

For each goal use its linked F workflow for trigger, preconditions, success path, alternate path and final state. Shared preconditions are authorized role/resource scope and required services; exceptions must preserve truthful state. UC02 requires valid code and current eligibility; UC03 requires publisher authority and current subscriber admission; UC06 requires approved network scope; UC08 requires end authority before mutation. UC11–UC13 include lower-priority capabilities and do not silently become Must work.

3. Example — UC02 Request and obtain admission

Trigger: student opens local join page. Preconditions: correct network and trusted endpoint; active class. Student enters code and optional agreed identifier. Backend validates code, class, block and capacity; if approval mode applies, wait without viewing rights. Owning teacher decision is reauthorized. Final atomic checks allocate a seat and issue permitted subscription credentials. Pending, rejected, expired, full or revoked requests receive no viewing grant. Postcondition: admitted/waiting or admitted/viewing, or a clear non-admitted result. F05 and TC-S02/TC-ADM01–07 specify acceptance.

4. Administration coverage

Account provisioning, role assignment and app settings are supporting proposals in Area 06. Keep them explicit in permissions/design; do not renumber UC01–UC15 to disguise them as source-defined use cases. Their scope needs D01/D14 approval.

Source: 02-system-workflows.md

System workflows F01–F18

Version: 0.1 · **Prepared:** 9 November 2026 · **Status:** Proposed review draft; no teacher approval or product test execution claimed.

1. Diagram authority and reading order

These are compact **proposed process overviews**, not executable code or proof that every concurrency path has been implemented. The original detailed Mermaid remains in REQUIREMENTS-AND-ACTIVITIES.md and the corrected F01 source in board/row-3/F01-CLASSROOM-LIFECYCLE.mmd. The business rules in this package capture the later reviewed separation of authorization, stale operations, closure and restoration. Do not reproduce a generated image as an exact rendering unless generated from the checked code.

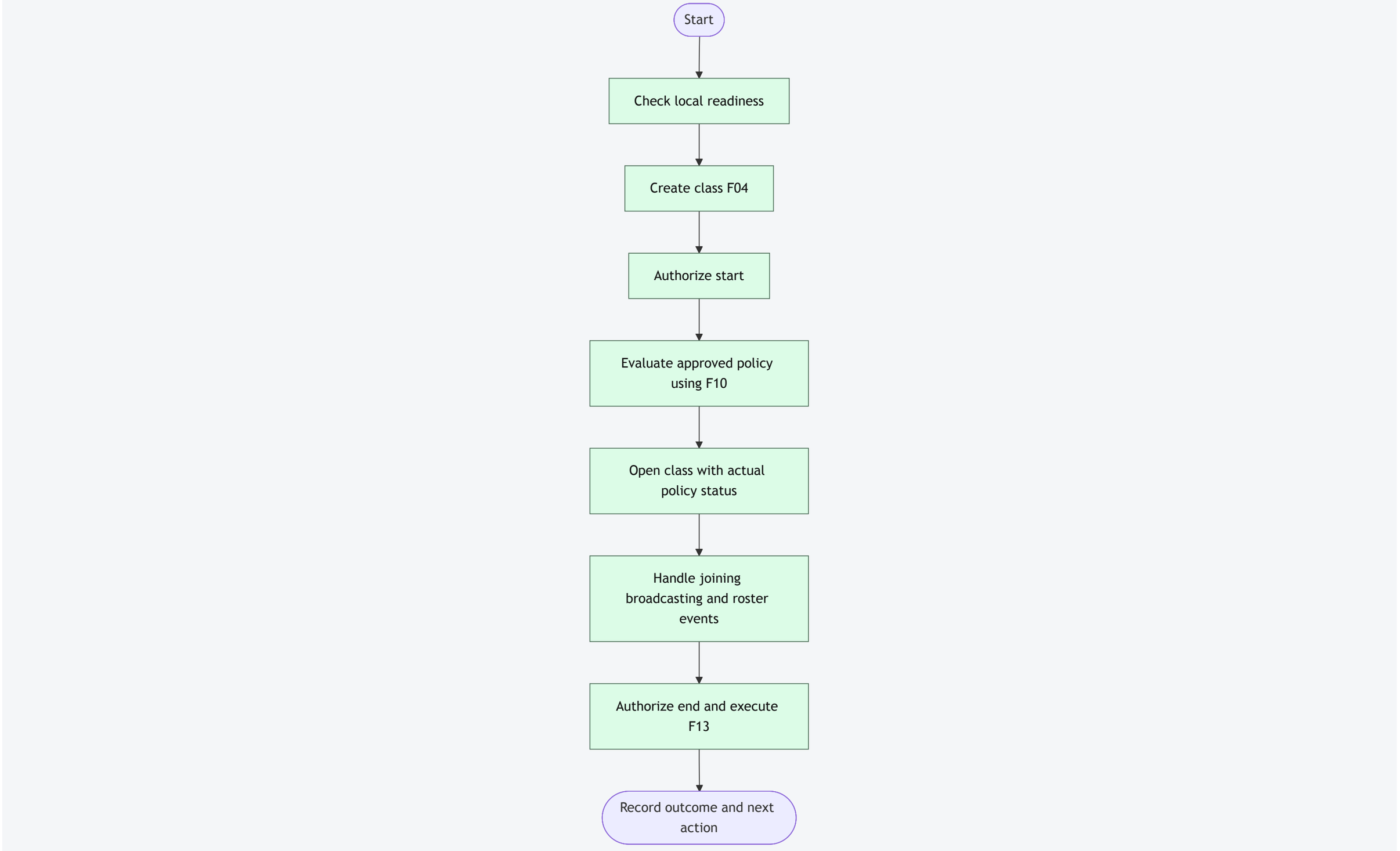
Suggested board review order: F01 → F05 → F06 → F10 → F13 → F08 → F07 → F12 → F15 → F14, then F04/F09/F11, deployment F02/F03 and selected F16–F18. This is a documentation review order, **not** mandatory runtime execution.

Green = Must; yellow = Should; gray = Could; red = proposed Won't; white = context/decision. CN/OR steps use proposed Must* classification. Exceptions remain mandatory when an optional capability is selected.

2. F01 — Entire classroom lifecycle

Trigger / actor: Prepared classroom; operator and teacher.
References: F04/F05/F06/F08/F10/F13; D05.

Diagram 1



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

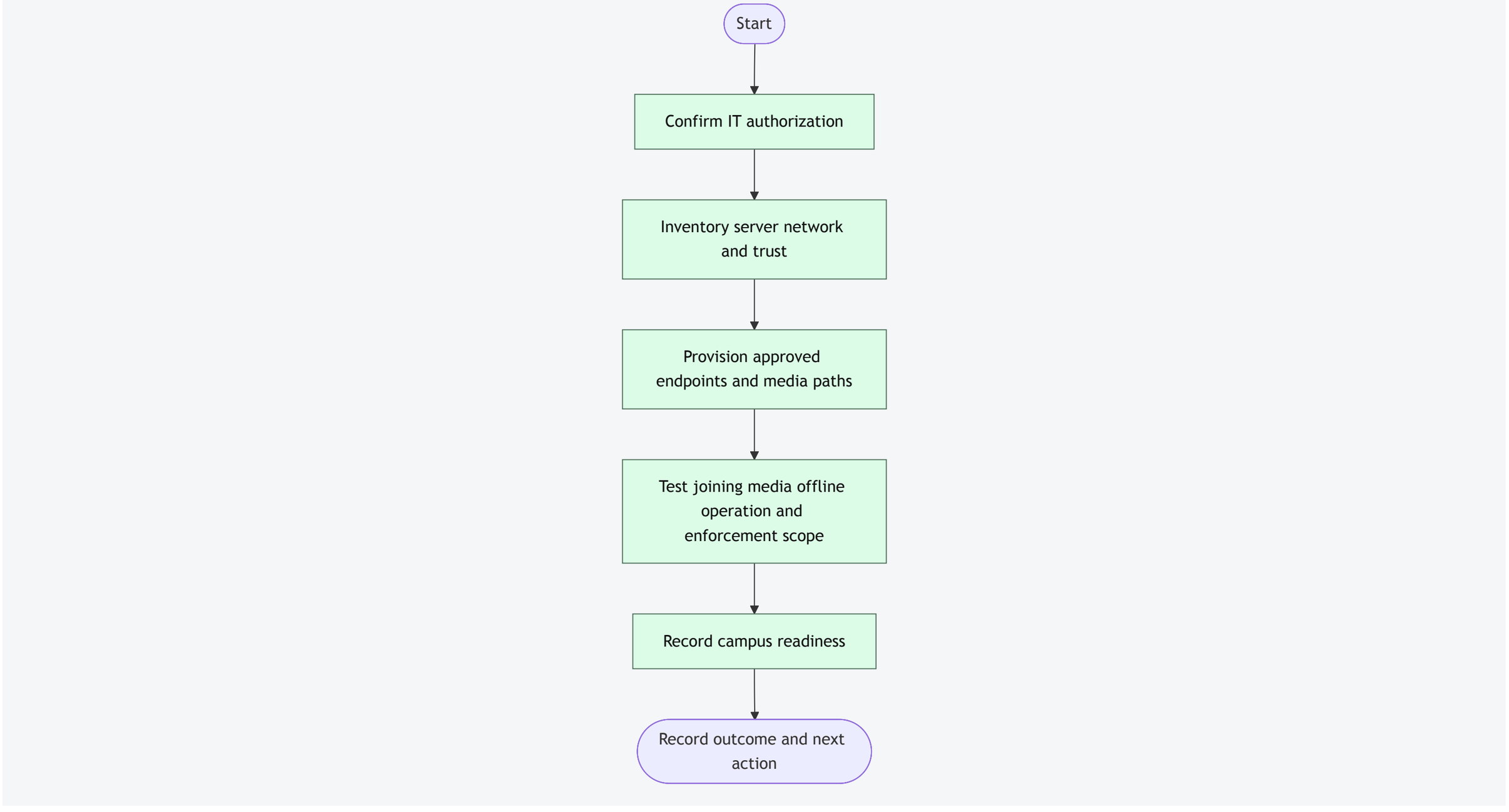
Exception and recovery: Unverified restrictions require the D05 agreed rule; aborted startup restores owned changes. Individual rejected joins do not end class.

Postcondition / next action: Class ended only after access/media closure verified; restoration is separate; F15 handles unresolved cleanup.

3. F02 — Prepare campus deployment

Trigger / actor: Authorized campus installer; before class.
References: CN01–CN10; D10/D12/D18.

Diagram 2



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

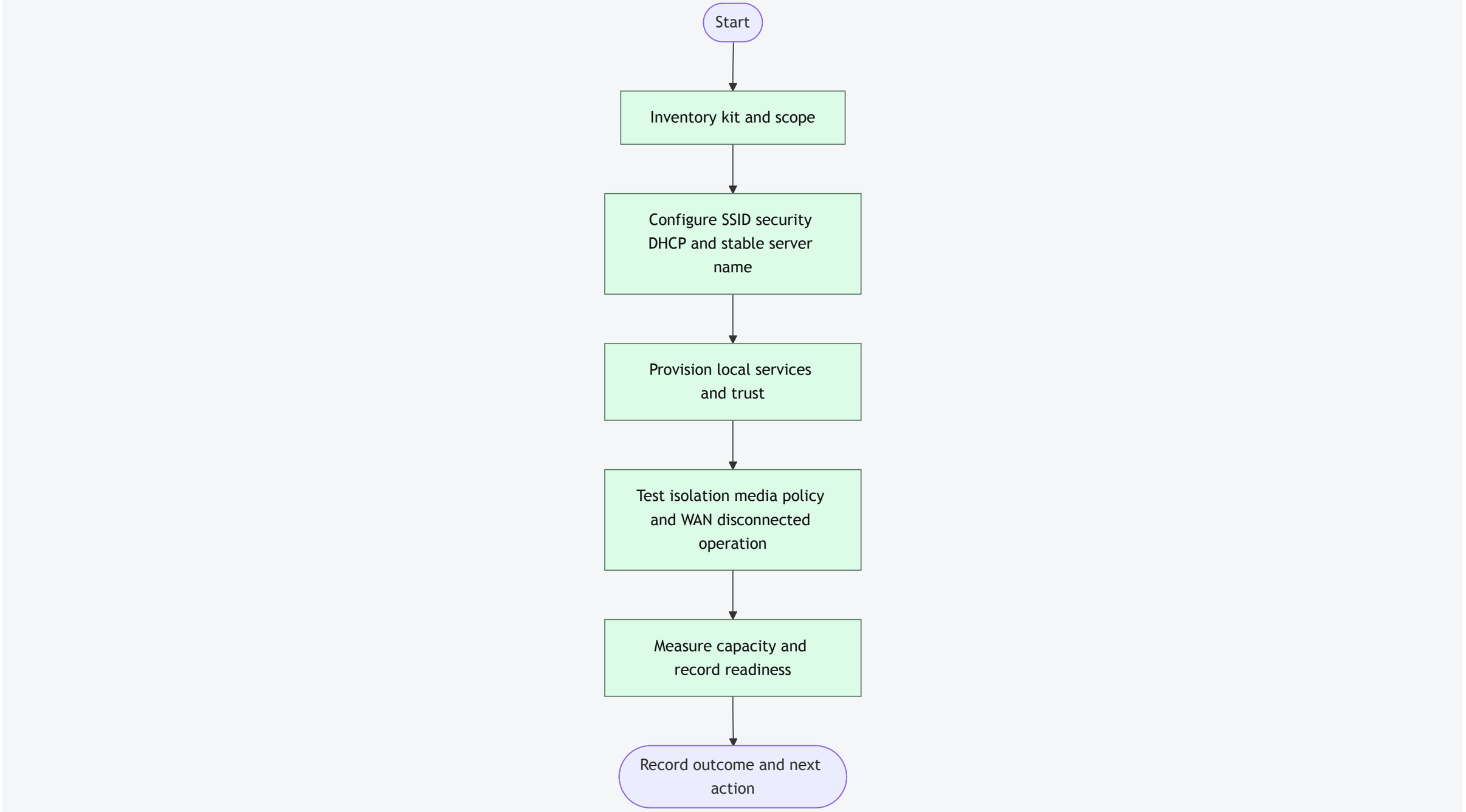
Exception and recovery: Missing permission, trust or media reachability prevents readiness; fix and retest.

Postcondition / next action: Prepared campus environment for F01; retain measured capacity and unverified integration gaps.

4. F03 — Prepare portable deployment

Trigger / actor: Authorized portable installer; before class.
References: OR01–OR12; D12/D18.

Diagram 3



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

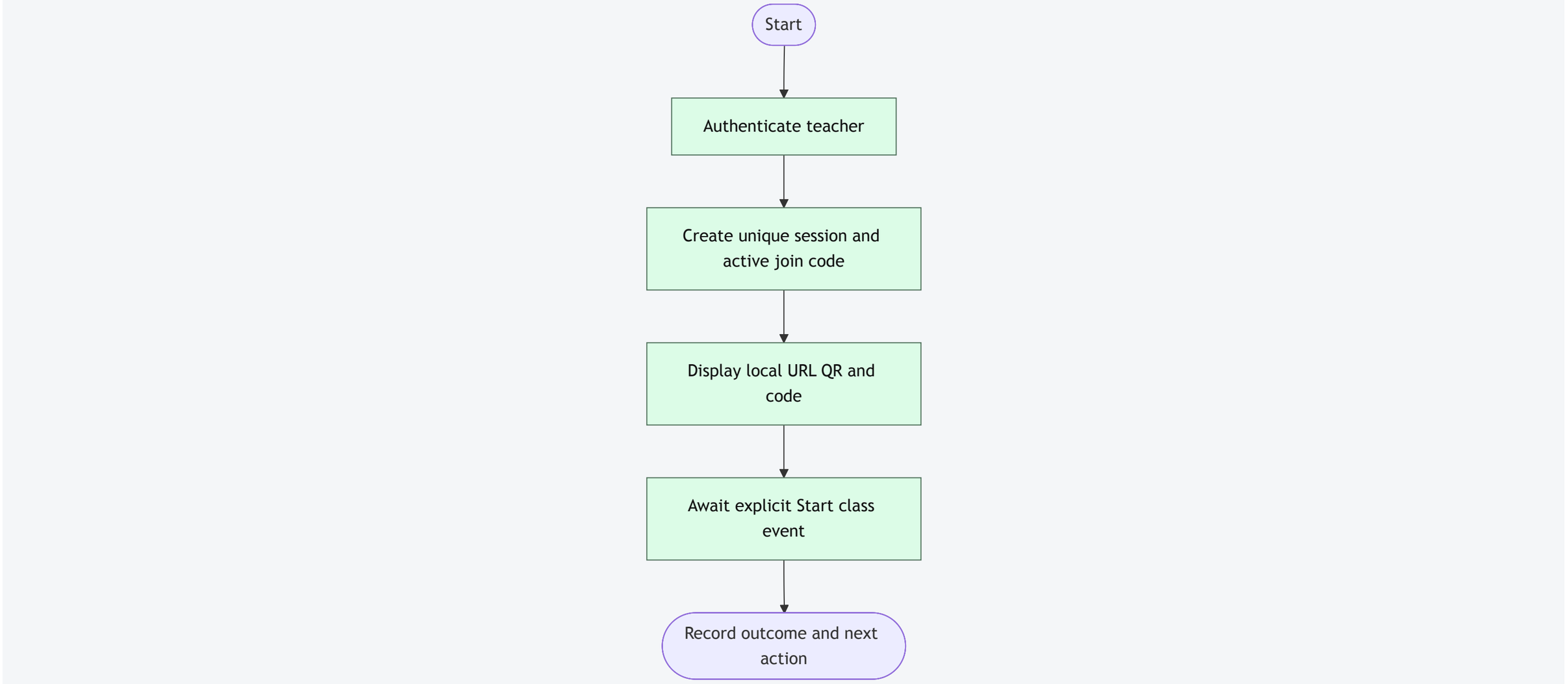
Exception and recovery: Insufficient AP capacity or unsupported adapter stays a documented gap; no per-class manual router-edit workaround counted as automation.

Postcondition / next action: Prepared portable environment for F01; equipment and recovery record retained.

5. F04 — Create classroom

Trigger / actor: Teacher requests new classroom.
References: T01/T08; D04.

Diagram 4



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

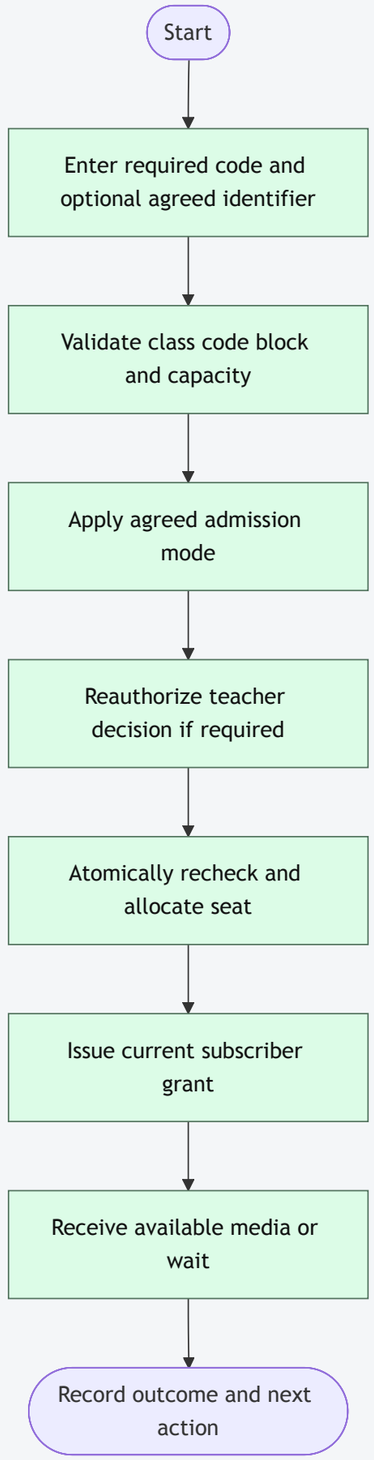
Exception and recovery: Unauthorized request creates no class; conflicting active code is regenerated safely.

Postcondition / next action: Created class; policy application and screen capture have not automatically happened.

6. F05 — Student joining and admission

Trigger / actor: Student opens local URL on classroom network.
References: S01/S02/S07/T05/T06; D02/D03.

Diagram 5



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

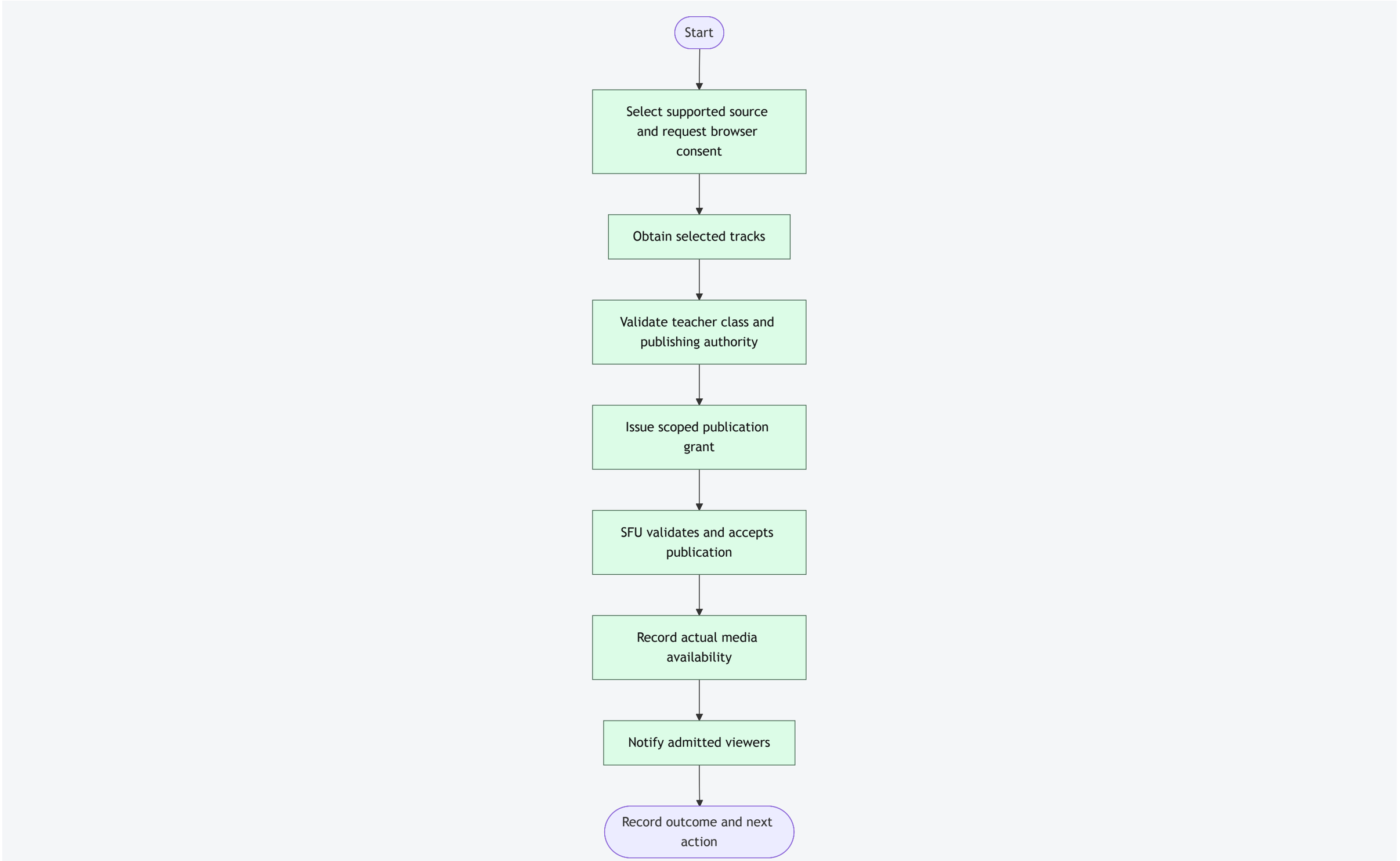
Exception and recovery: Pending/rejected/full/closed requests have no viewing grant. Recheck capacity at commit; rejected unauthorized teacher action does not reject the student.

Postcondition / next action: Admitted waiting or viewing; F12 reconnects, F08 manages access, F13 ends class.

7. F06 — Start screen and audio broadcasting

Trigger / actor: Authorized teacher starts or retries publication.
References: T02/T03/T04/T07/S03/S04; D09.

Diagram 6



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

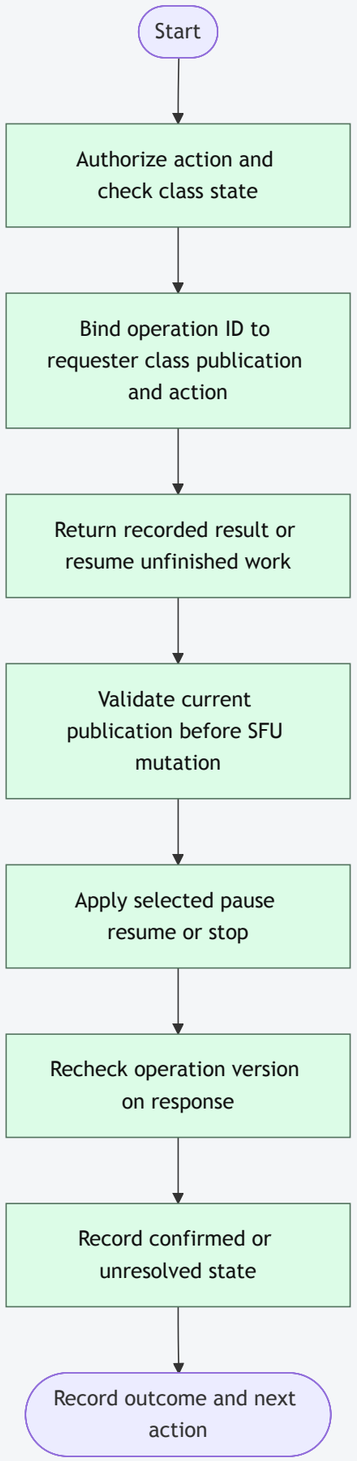
Exception and recovery: Capture refusal produces no publication. Missing optional mic must not be reported as captured. Service/authorization failure stays not publishing; release captured tracks when denied.

Postcondition / next action: Broadcast available even with zero viewers; F07 controls publication; F13 closes class.

8. F07 — Pause resume and stop broadcasting

Trigger / actor: Teacher controls current publication.
References: T02/T07/S03/S04/S05; D08.

Diagram 7



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

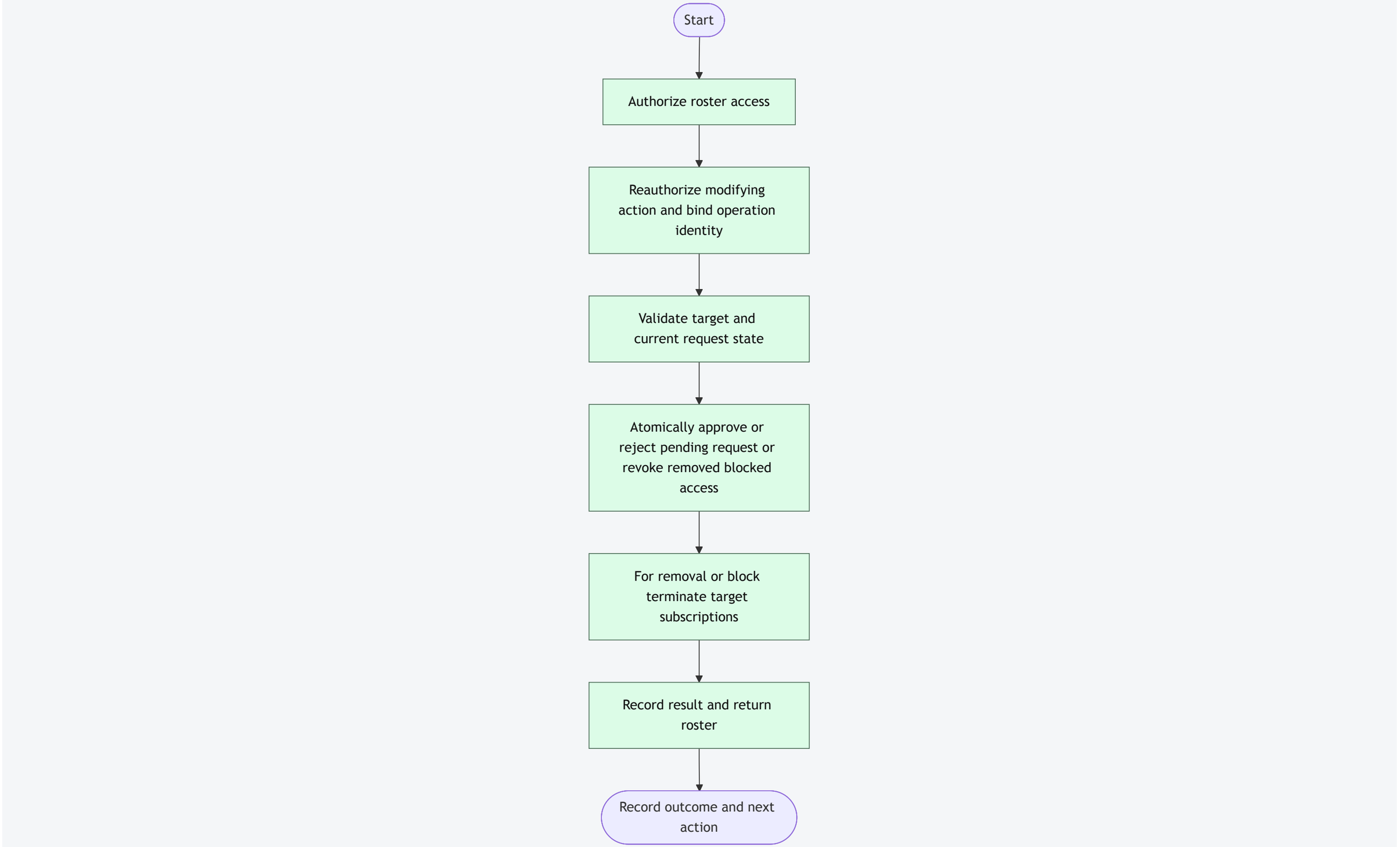
Exception and recovery: Closure takes precedence via F13; stale publication response cannot mutate new stream. Resume with ended capture goes to F06; failures route F12/F15.

Postcondition / next action: Pause/resume/stop leave admission intact. Stop ends publication, not class. D08 pause/audio semantics remain pending.

9. F08 — Participant management

Trigger / actor: Teacher opens roster and selects an action.
References: T05/T06; D02/D03.

Diagram 8



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

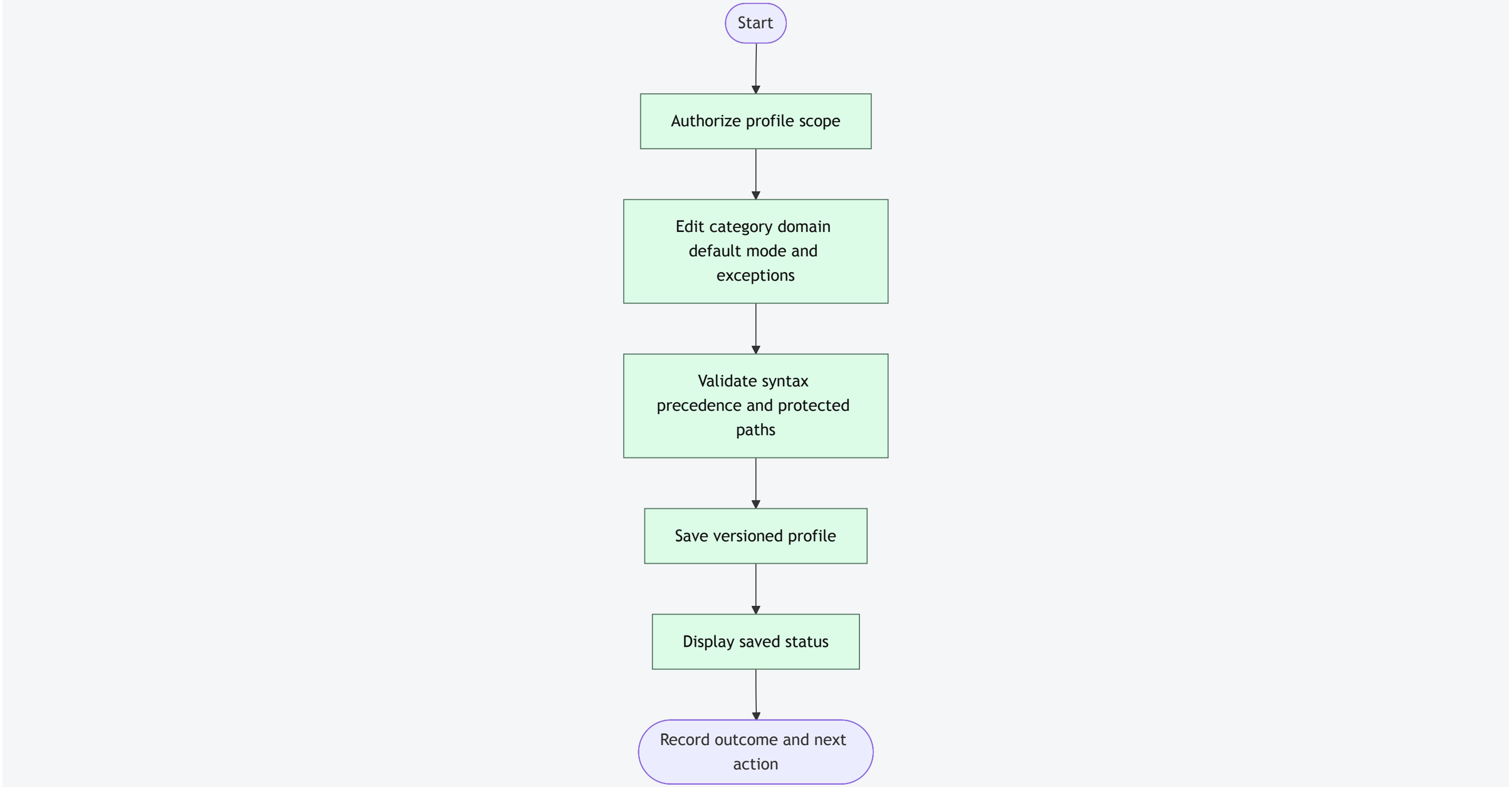
Exception and recovery: Approve rechecks final seat; rejected action changes nothing. SFU failure means removal recorded but media cleanup unresolved; schedule F15.

Postcondition / next action: Class stays open and teacher publication continues. Approved student continues F05; display names are not durable block identities.

10. F09 — Create or edit restriction profile

Trigger / actor: Scoped SYS or NET edits configuration.
References: NAC02/03/04/07/08/16/17/18; D07/D15.

Diagram 9



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

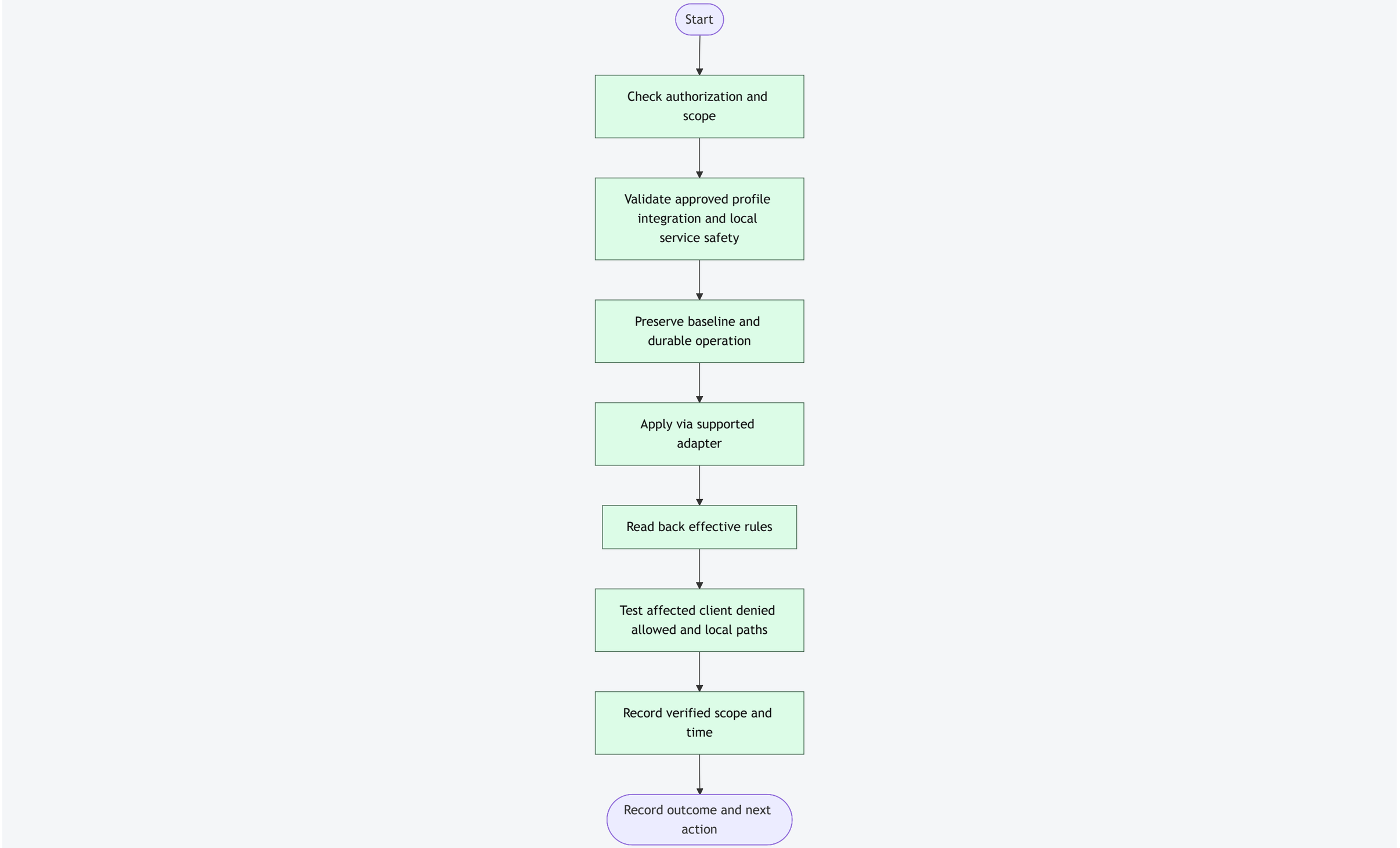
Exception and recovery: Invalid or out-of-scope edit is denied; no network mutation from save.

Postcondition / next action: Saved profile eligible for separately authorized F10 application.

11. F10 — Apply and verify restrictions

Trigger / actor: Authorized manual request or approved class-start automation.
References: NAC01/05/10/11/16/17/18/19; D05/D06/D07/D15.

Diagram 10



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

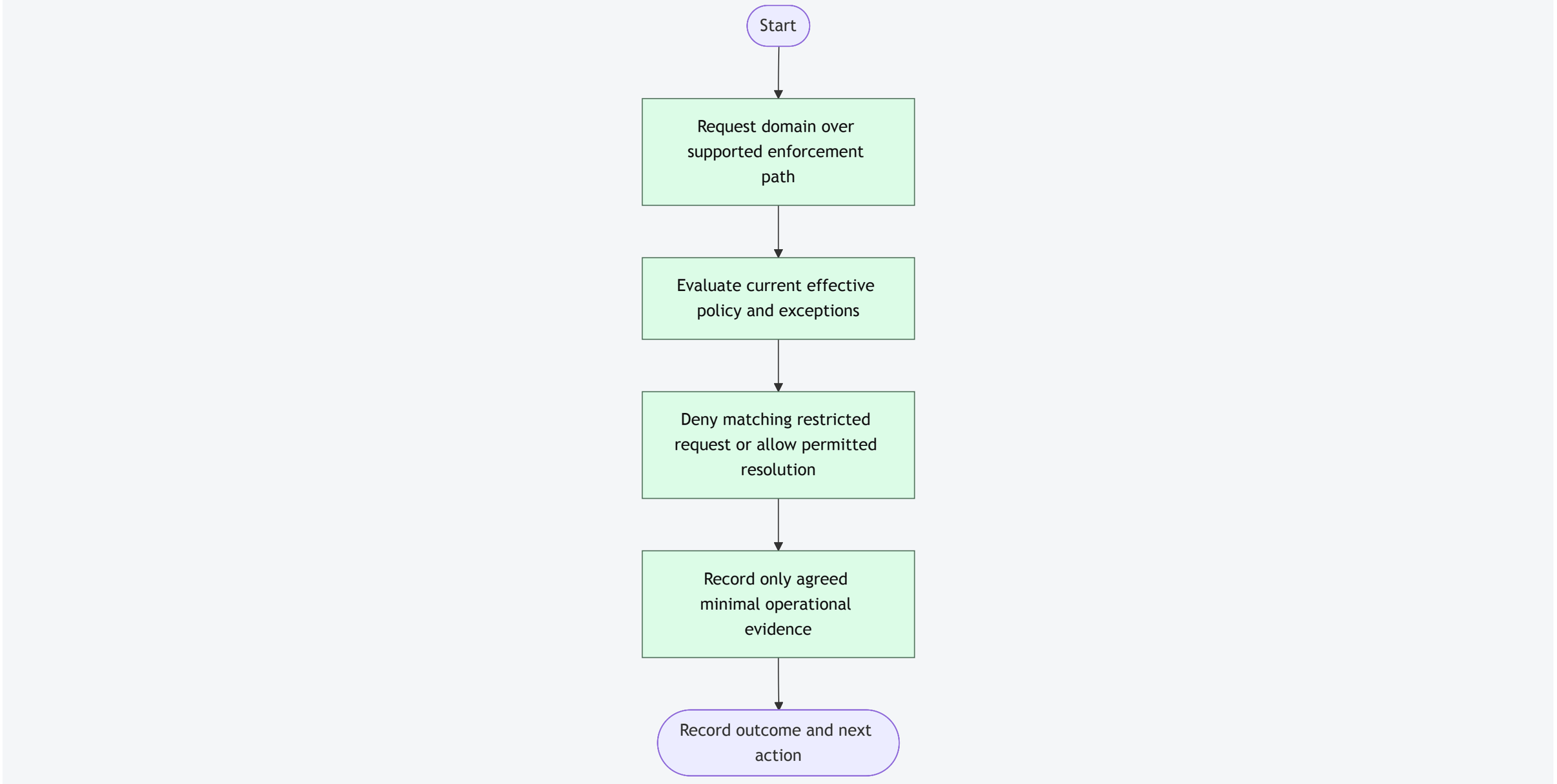
Exception and recovery: Rejected/partial/unknown result requires inspect/reconcile. Restore owned changes if needed; verified restoration or unresolved recovery is recorded.

Postcondition / next action: Return actual result to F01; F10 does not decide whether class may continue under D05.

12. F11 — Public website request

Trigger / actor: Student device requests an external domain.
References: NAC02/03/04/08/18; D07.

Diagram 11



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

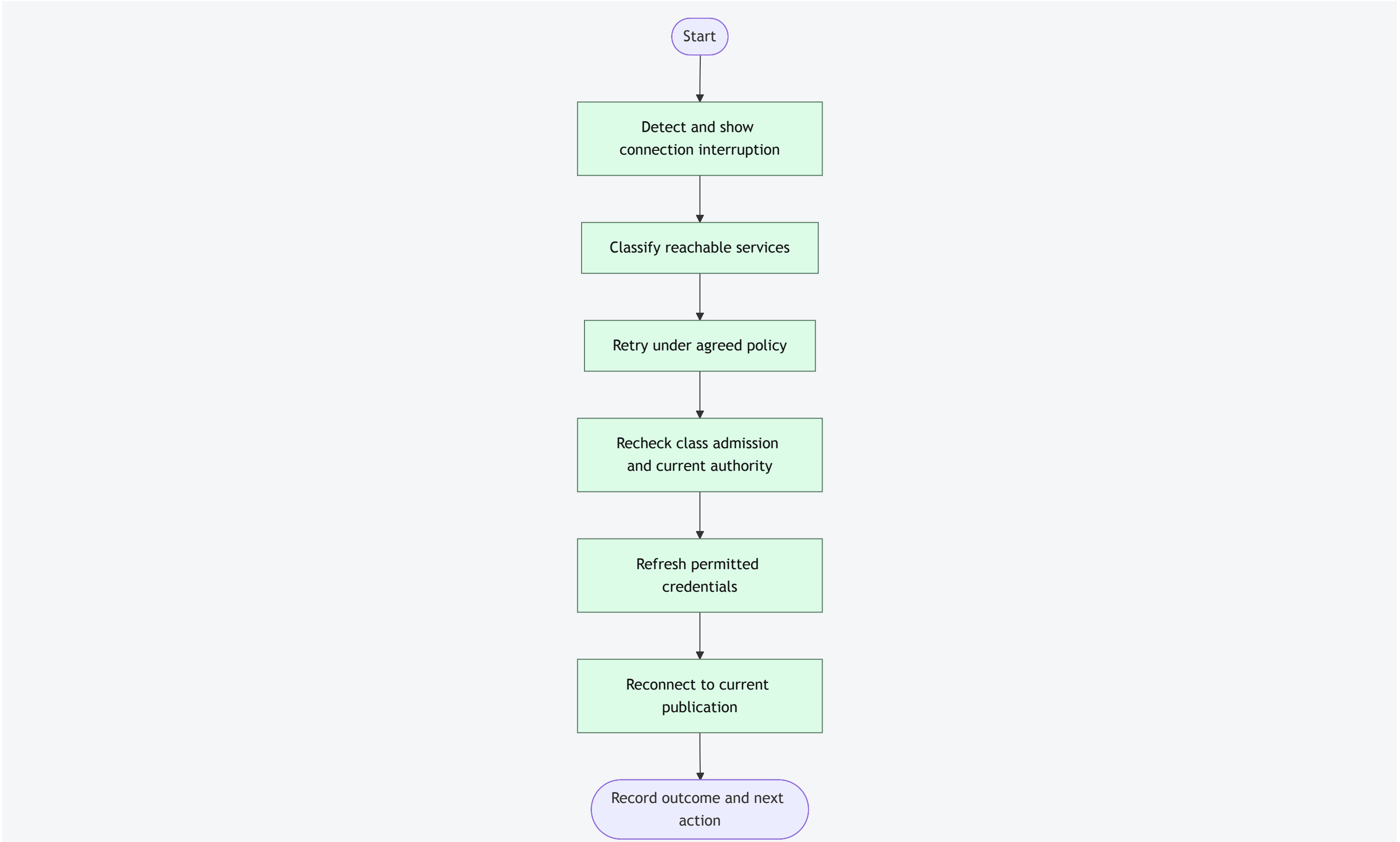
Exception and recovery: Alternate DNS/VPN/direct IP may bypass domain-only controls; WAN outage is not proof of selective denial.

Postcondition / next action: Local classroom remains available; show restrictions only for tested supported scope.

13. F12 — Connection loss and reconnection

Trigger / actor: Student or teacher media connection is interrupted.
References: S06/S07; D11.

Diagram 12



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

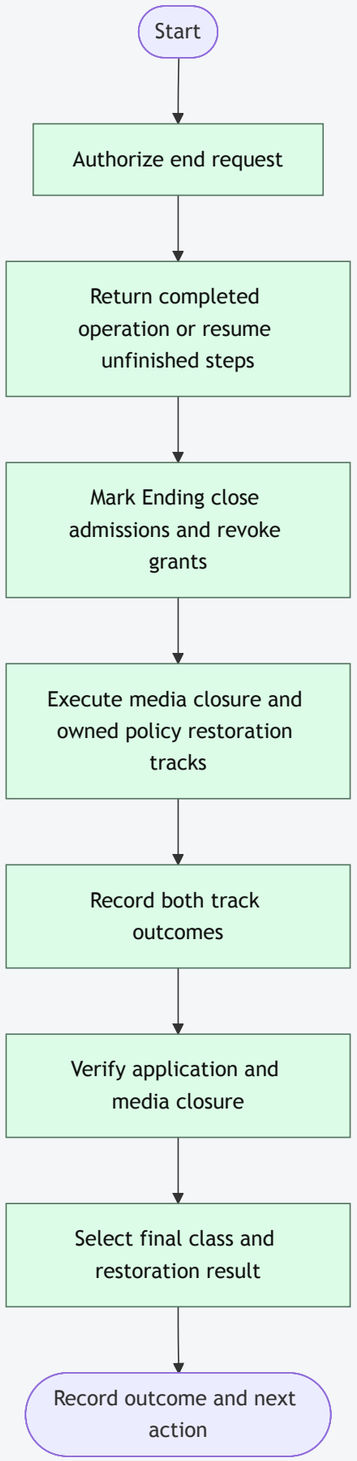
Exception and recovery: Ended class or revoked admission denies reconnect; lost capture requires explicit F06 consent. Persistent service failure routes F15.

Postcondition / next action: Same authorized class resumes or waiting/unresolved/denied status is shown; no automatic re-admission.

14. F13 — End class and restore policy

Trigger / actor: Teacher requests End class.
References: NAC06/10/11; D11.

Diagram 13



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

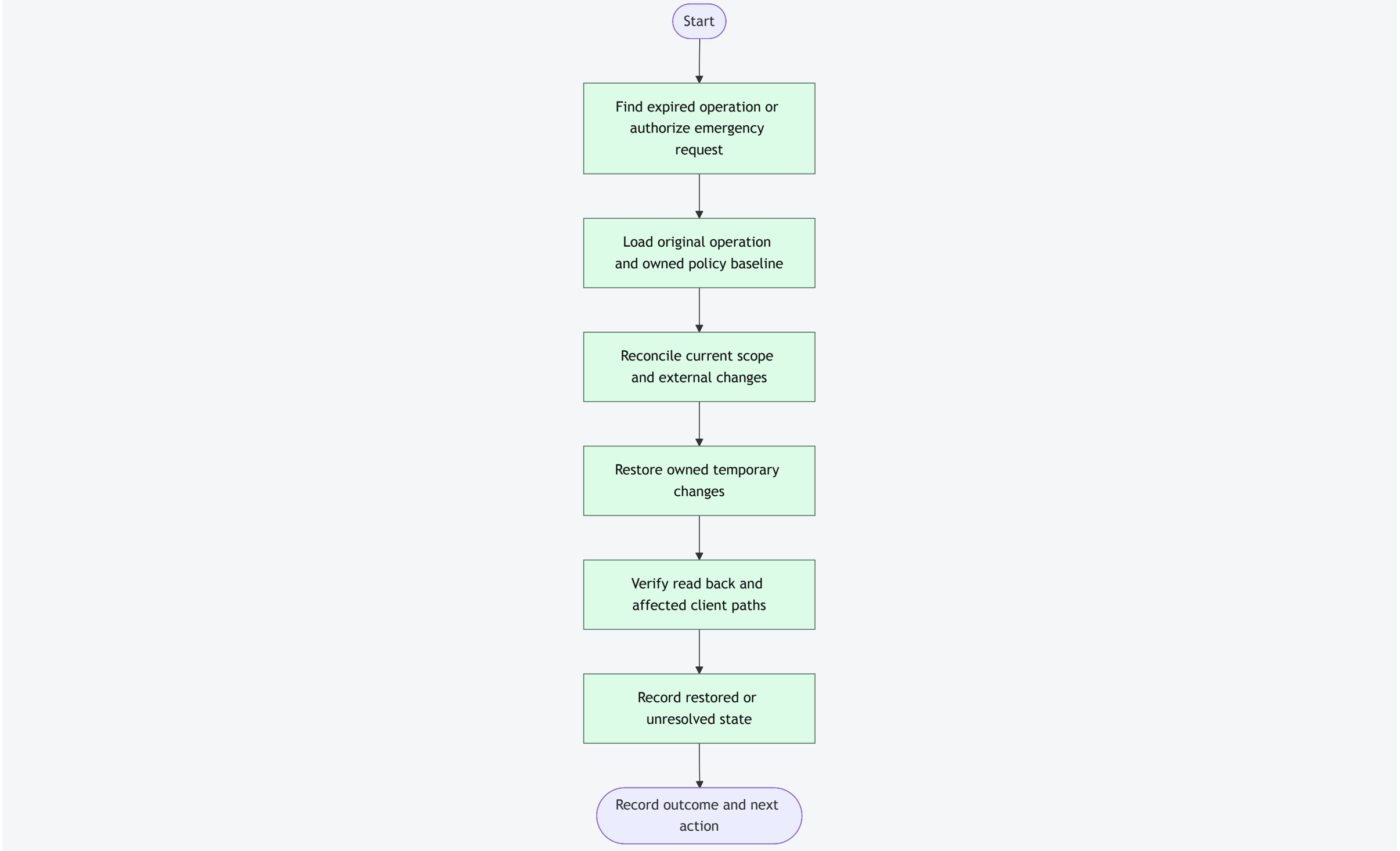
Exception and recovery: Unauthorized request leaves class unchanged. Closure unverified means Class ending; verified closure can coexist with pending restoration. Conflicting external network changes require reconciliation.

Postcondition / next action: Class ended with restoration not required verified or pending; otherwise Class ending cleanup unresolved. F15 continues recovery.

15. F14 — Temporary expiry and emergency disable

Trigger / actor: Durable expiry trigger or scoped authorized emergency action.
References: NAC12/NAC13; D11/D15.

Diagram 14



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

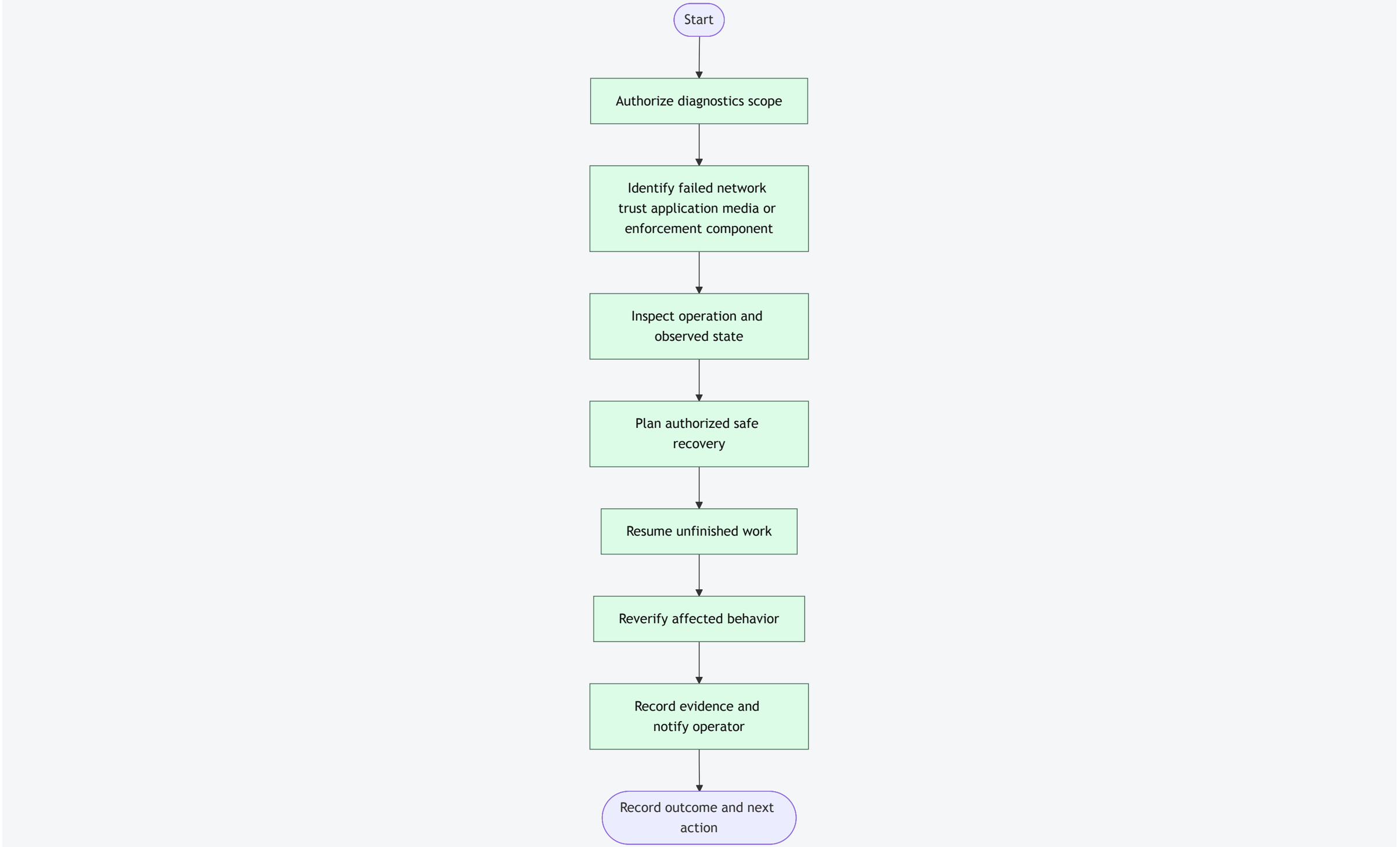
Exception and recovery: Unreachable or conflicting state is persistent unresolved recovery, not disabled/restored; repeated trigger is idempotent.

Postcondition / next action: Return status to active classroom/operator; F15 handles unresolved recovery; emergency policy action does not itself end class.

16. F15 — Diagnostics and recovery

Trigger / actor: Authorized operator or durable unresolved operation.
References: Shared operational obligations; T09 Should; D11.

Diagram 15



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

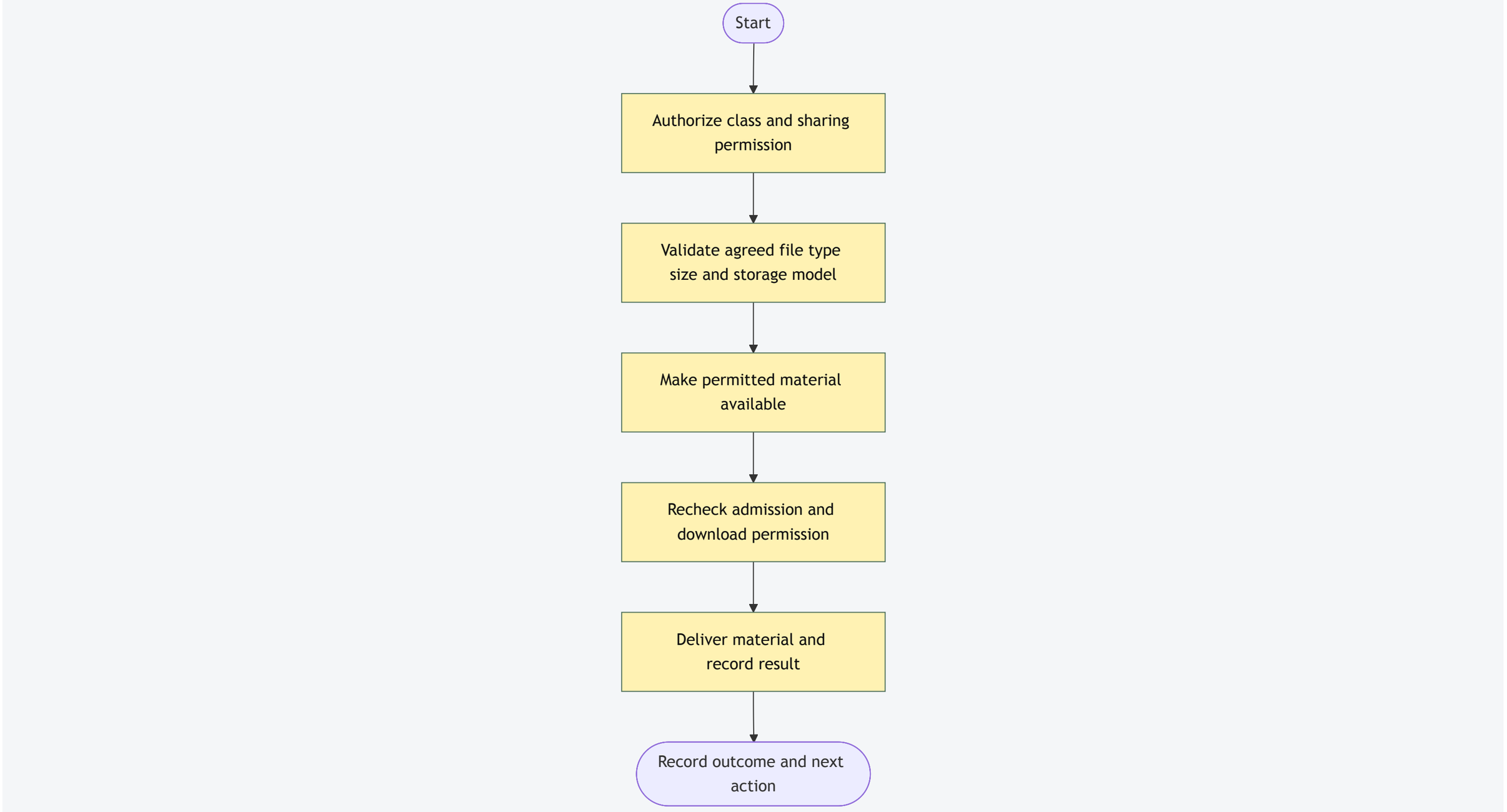
Exception and recovery: Never replay completed mutation or overwrite unrelated network changes; persistent fault remains unresolved and visible.

Postcondition / next action: Return verified result to originating flow or durable unresolved record with next operator action.

17. F16 — Share and download materials

Trigger / actor: Teacher shares approved materials; admitted student requests them.
References: T10/S09 Should; D16.

Diagram 16



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

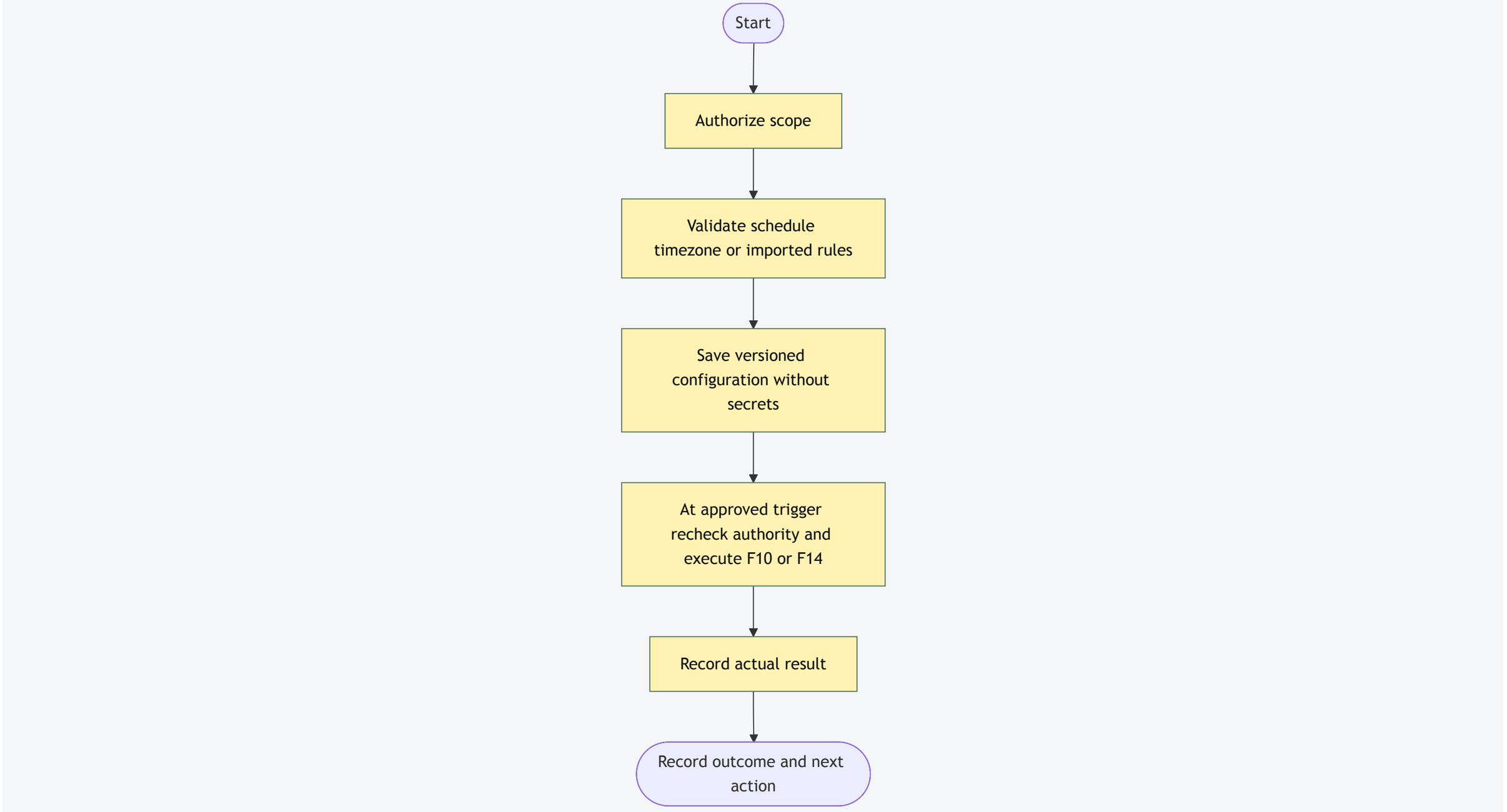
Exception and recovery: Reject forbidden type/size or revoked access; direct file route cannot bypass authorization.

Postcondition / next action: Material accessed or clear denial; class remains open; content handling details await D16.

18. F17 — Schedules and configuration transfer

Trigger / actor: Scoped administrator manages policy tools.
References: NAC09/14/20 Should; D11.

Diagram 17



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

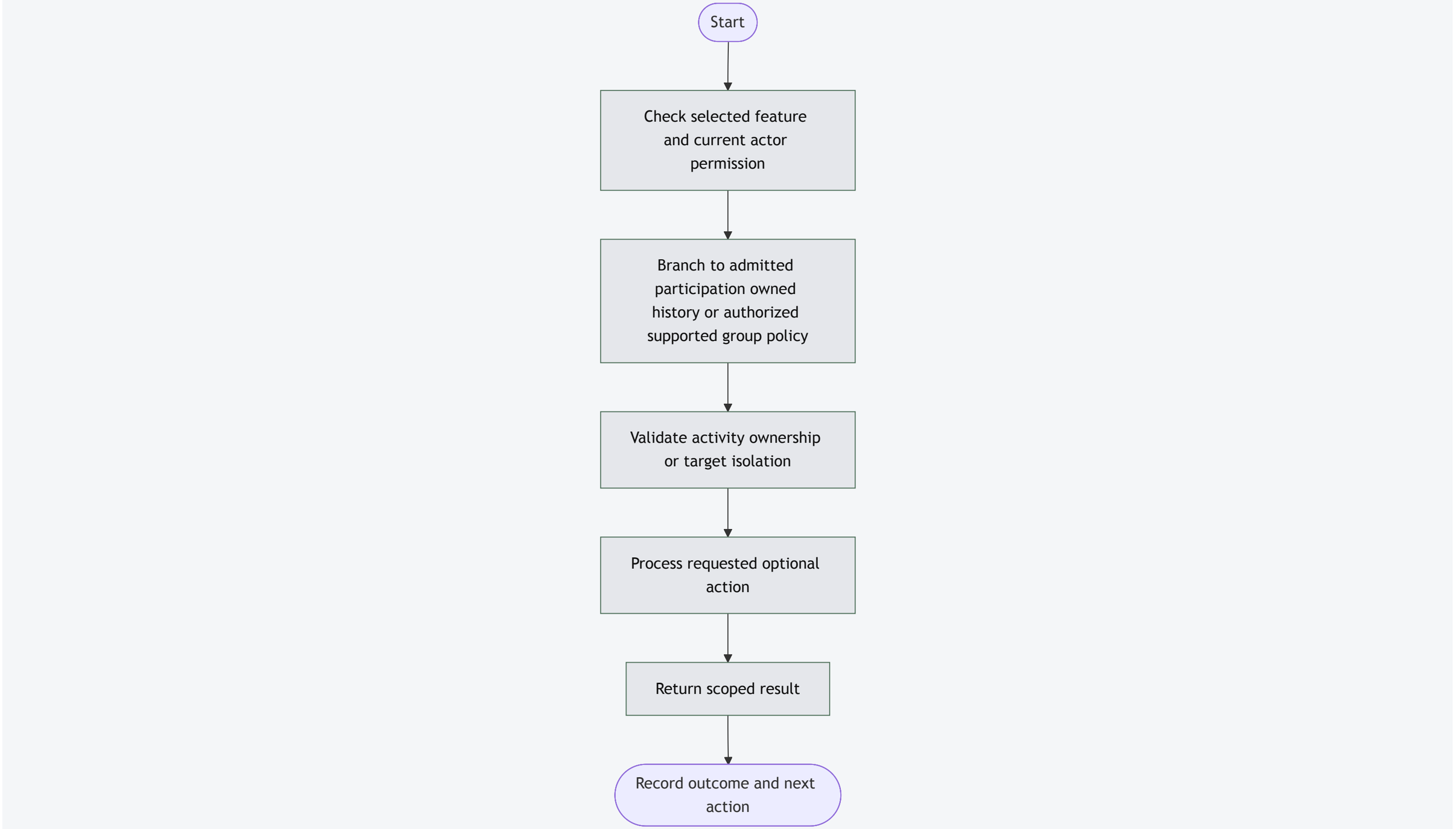
Exception and recovery: Malformed import changes nothing; missed schedule/retry behavior needs D11. Required offline runtime remains mandatory.

Postcondition / next action: Configuration saved or scheduled operation recorded; applying and verification remain separate.

19. F18 — Optional participation history and group policy

Trigger / actor: Actor uses an explicitly selected optional feature.
References: S10/T11/NAC15 Could; D17.

Diagram 18



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

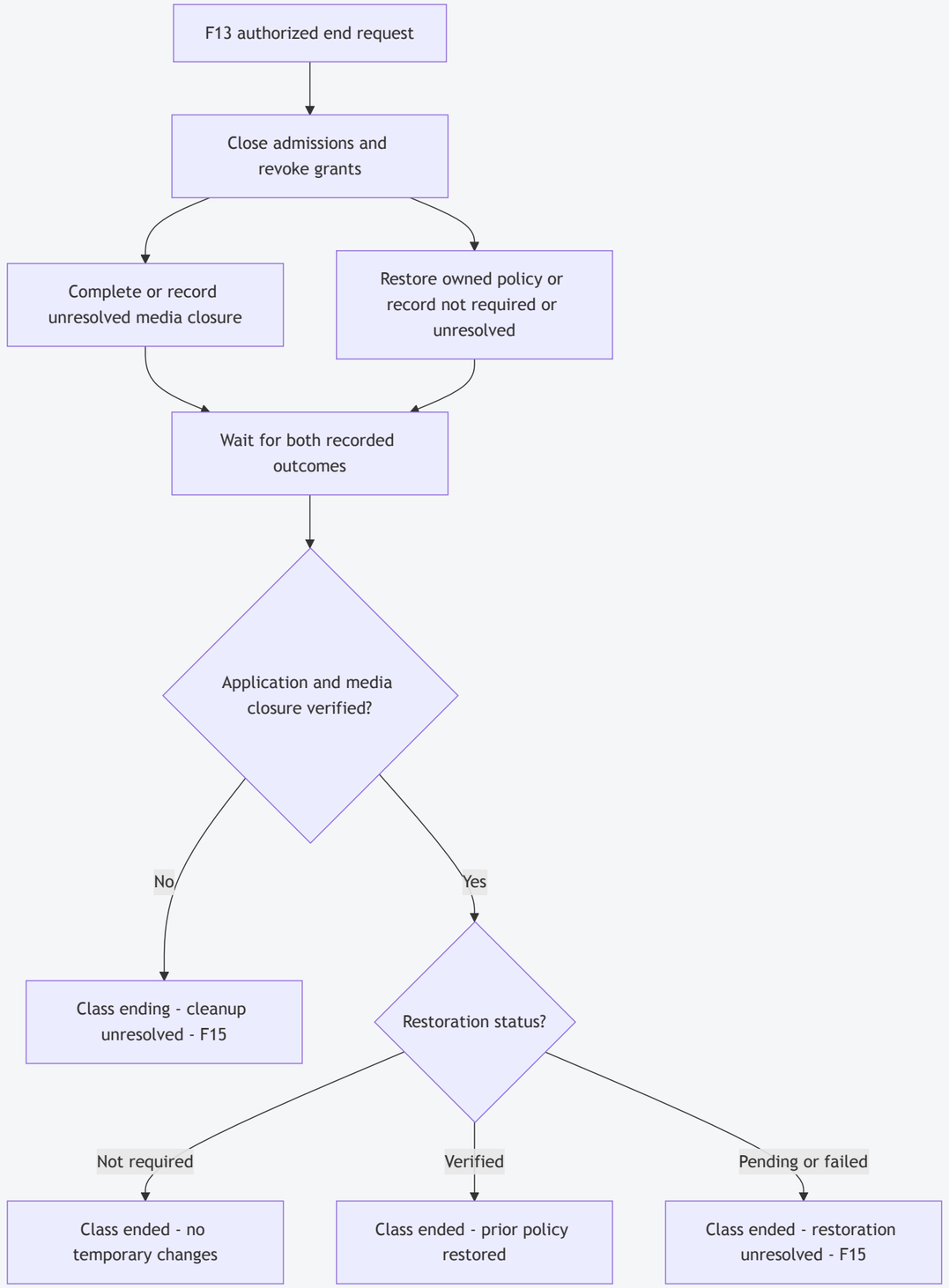
Exception and recovery: Unselected feature is not required; unauthorized submissions/history access denied; unsupported group identity cannot be claimed enforced.

Postcondition / next action: Return to class or admin view; no implied recording or remote student monitoring.

20. Mandatory branch diagrams for high-risk transitions

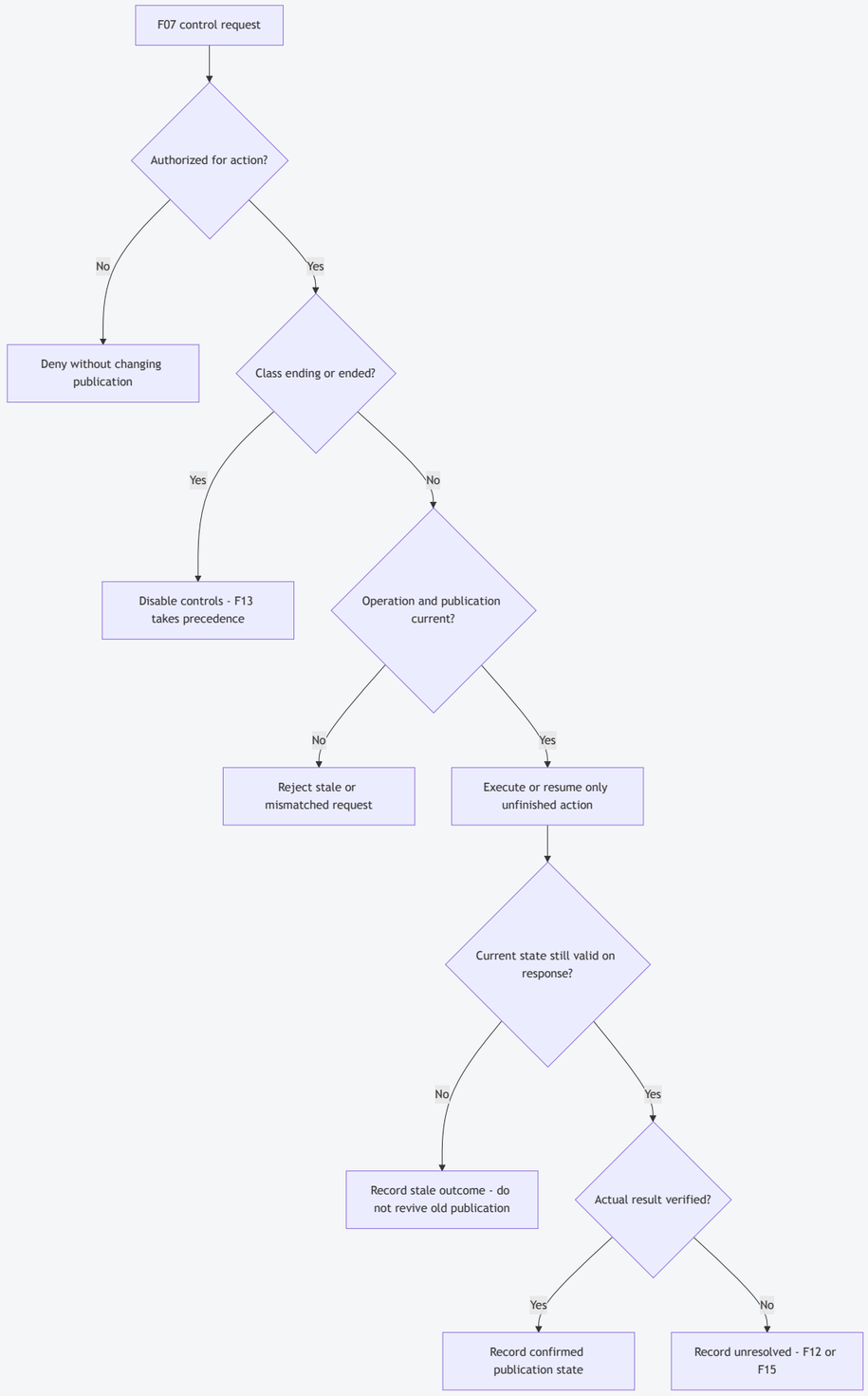
The overviews above show the principal sequence; the following diamonds define essential branching that must not be lost in a presentation.

Diagram 19



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

Diagram 20



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

Detailed inherited activity diagrams

Flow 01 — Entire classroom lifecycle

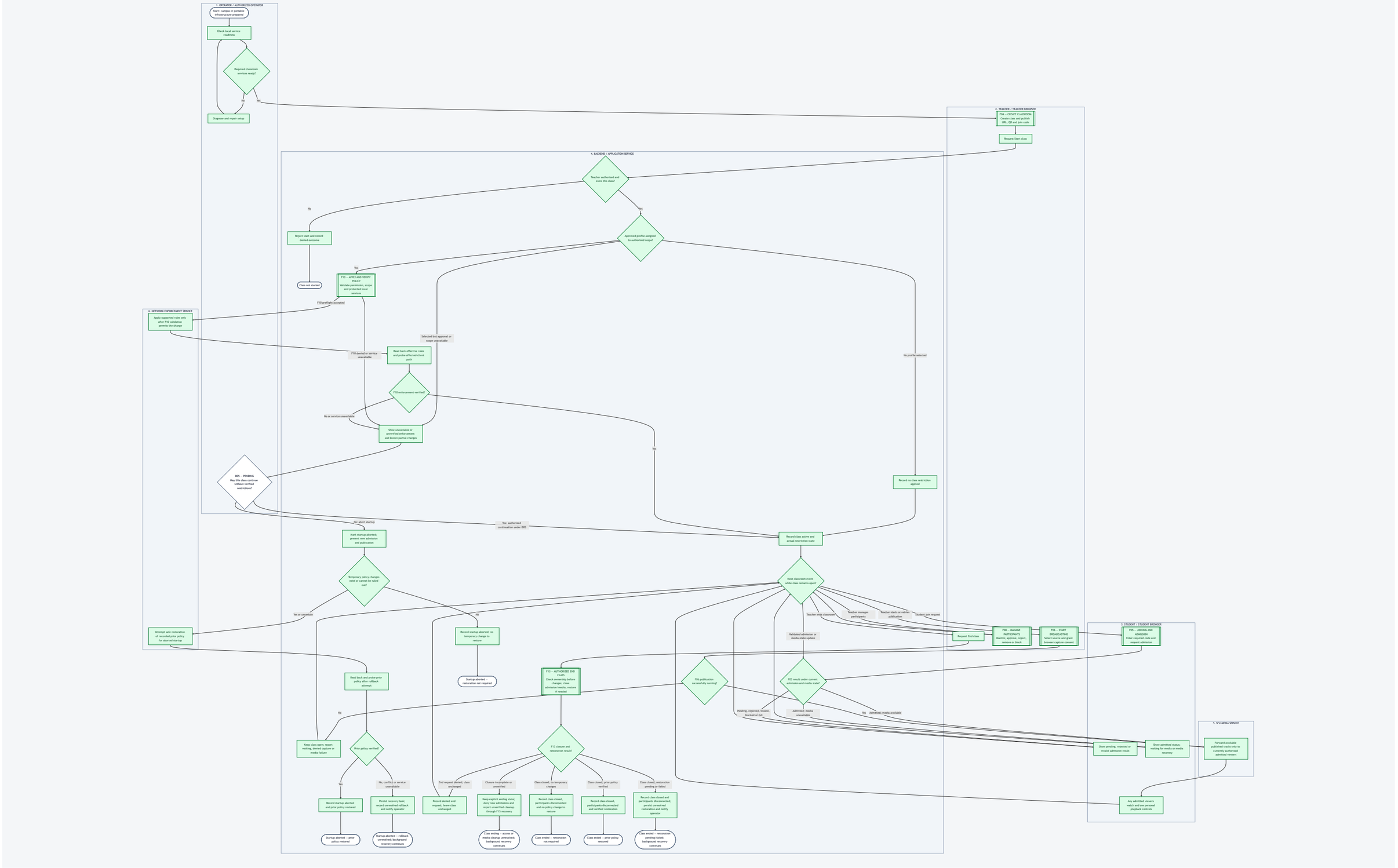
Responsibility lanes: neutral backgrounds identify Operator, Teacher/browser, Student/browser, Backend, SFU and Enforcement Service. Individual node colors retain MoSCoW meaning; diamonds are conditions and double-bordered boxes reference detailed workflows.

Workflow classification: MUST — overview of required capabilities. D05 remains a white, unresolved client/operator choice; no proposed continuation rule is treated as approved.

Trigger: A prepared campus or portable classroom is about to begin. **Outcomes:** unauthorized start denied; startup aborted with distinct rollback outcomes; or a running class ends with restoration not required, verified or unresolved. If access/media closure itself is not verified, show **Class ending — cleanup unresolved**, not Class ended.

Overview semantics: F01 coordinates the detailed processes; it does not replace F04, F05, F06, F08, F10 or F13. The classroom-event loop is a compact view of repeated/asynchronous events, not a single-threaded implementation requirement. Student joining, pending-request completion, publication and participant control can occur independently while the class is open. Teacher publication does not wait for a first admission. An individual invalid/pending/rejected join does not terminate the class. Only current admitted viewers receive available media; successful publication with zero viewers is still valid.

Diagram 21



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

F01 subprocess references and pending decision

Reference	Responsibility in F01	Contract to preserve
F04 — Create classroom	Teacher initiates; Backend validates/stores class	This overview proceeds after successful authorized creation. Creation/validation/media-setup errors remain in F04 and do not constitute successful startup
F05 — Joining and admission	Student submits code; Backend validates; owning Teacher decides where configured	Return actual admission/media state. Pending and rejected requests get no viewing grant. Recheck identity, class/code, block and capacity before allocating access
F06 — Start broadcasting	Teacher/browser capture and Backend/SFU publication authorization	Return publication running or not running. Permission denial/cancellation, unauthorized publish and media failure leave the class open with truthful waiting/error status
F08 — Participant management	Owning Teacher and Backend/SFU	Enforce ownership and current access; removal/blocking revokes relevant media/admission. Approval completion may produce a later validated admission-state event
F10 — Apply/verify policy	Backend coordinates; Enforcement Service applies permitted rules	The labeled F10 node and its downstream service/read-back actions are parts of one policy operation, not two applications. Reject unauthorized/unsafe scope before mutation; keep last actual state and partial changes visible
F13 — Authorized end class	Backend validates request before closure; SFU/enforcement perform required cleanup	Denied end request leaves the class unchanged. Only completed access/media closure produces Class ended. Policy restoration is conditional and separately verified
F15 — Diagnostics/recovery	Authorized operator and service recovery	Track unverified access/media cleanup separately from network-policy restoration; do not claim closure because the request was submitted
D05 — Continue without verified restrictions?	Authorized operator under client-agreed rule	Still Pending. A normal lesson/exam may need different behavior; this diagram does not authorize a particular override

F01 final states and next operational action

Final state	Required record before the endpoint	Next action after the user-facing endpoint
Class not started — unauthorized start	Denied action and unchanged class state	Correct access or let the owning Teacher submit an authorized request

Final state	Required record before the endpoint	Next action after the user-facing endpoint
Startup aborted — restoration not required	No temporary policy mutation and admission/publication disabled for the aborted start	Resolve startup issue and make a fresh authorized start attempt
Startup aborted — prior policy restored	Verified rollback of temporary/partial policy changes	Resolve startup issue, revalidate scope and retry startup
Startup aborted — rollback unresolved	Durable recovery task, known state and operator notification	Retry safely/reconcile external changes; keep the scope unavailable for conflicting operations until recovered
Class ending — access/media cleanup unresolved	Explicit ending state, admission denial and unverified cleanup/recovery record	F15/operator completes and verifies cleanup; do not silently present the class as ended
Class ended — restoration not required	Completed access/media closure; no temporary change to restore	Return to dashboard, review permitted records or create a new class
Class ended — prior policy restored	Completed closure plus read-back/client-path verification of the prior policy	Return to dashboard, monitor services or create a new class
Class ended — restoration pending/failed	Completed access/media closure plus durable unresolved policy-restoration task	Background/operator recovery continues; prevent conflicting scope operations and update actual status when verified

Startup rollback and normal class-end restoration use separate branches and records. They may reuse safe adapter/recovery functions, but their lifecycle outcomes must not be merged. Restoration is an attempt until read-back and affected-path checks confirm it; unreachable/conflicting service state remains unresolved. Outcome recording occurs before each terminal node, which has no outgoing process edge. Background recovery after an endpoint is described in this table rather than drawn as an outgoing edge from a terminal.

Shape/color note: All six lane styles are explicitly defined in the code. Mermaid may reposition subgraphs; a reference/layout change must not alter permissions, branching or recorded outcomes. This is an unexecuted workflow specification, not evidence of successful recovery or client acceptance.

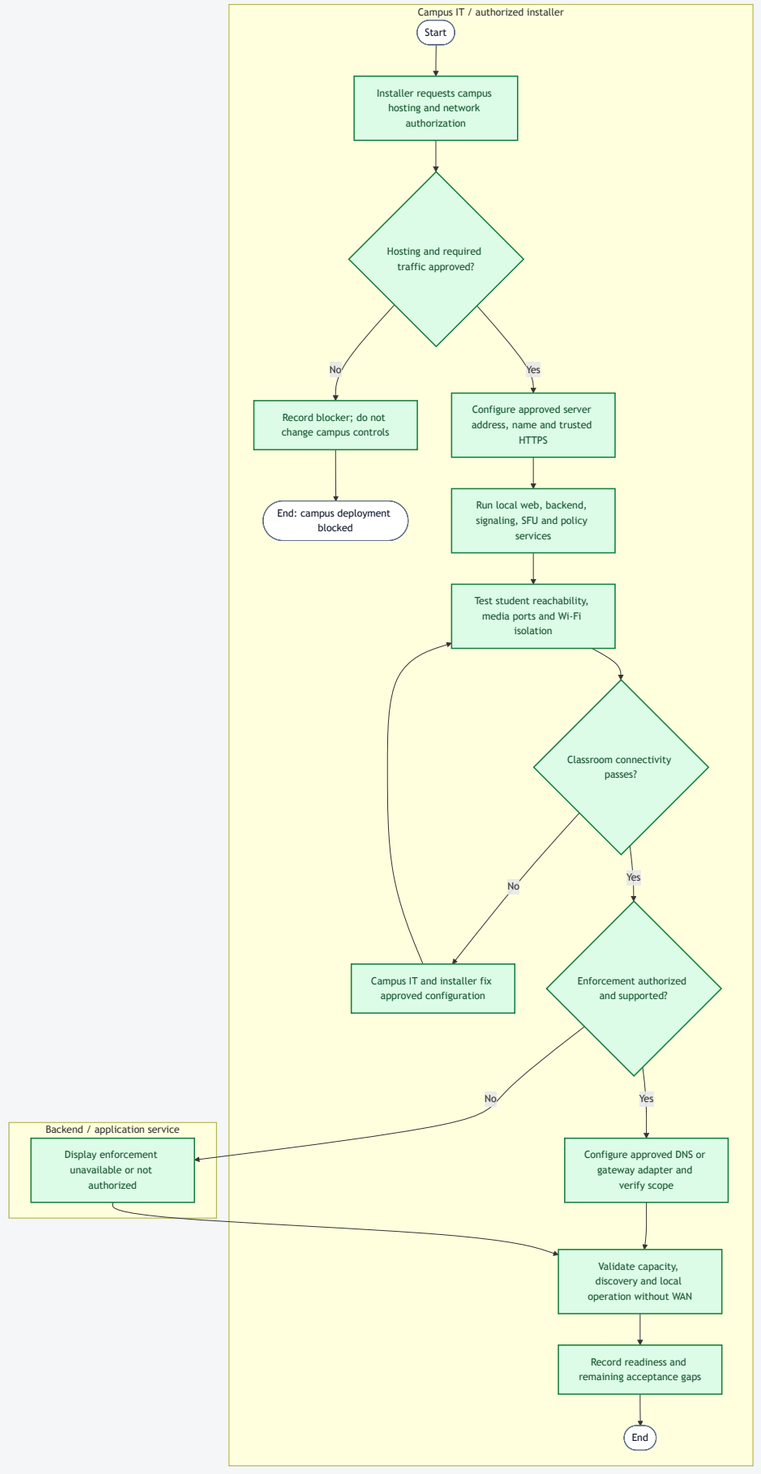
Flow 02 — Prepare campus deployment

Responsibility lanes: user/operator actions and automatic application/media/network actions are grouped separately. Authorization checks are server-side; delegated class automation uses recorded scope approval.

Workflow classification: MUST* — proposed MoSCoW label for source deployment obligations.

Covers CN01–CN10. This is installation/network preparation, generally performed before classes. Teachers should not need to edit router rules for every lesson.

Diagram 22



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

Campus browsing restrictions remain an acceptance gap if no approved integration exists. The system can expose that limitation accurately while classroom connectivity is tested; this does not mean CN10 is satisfied.

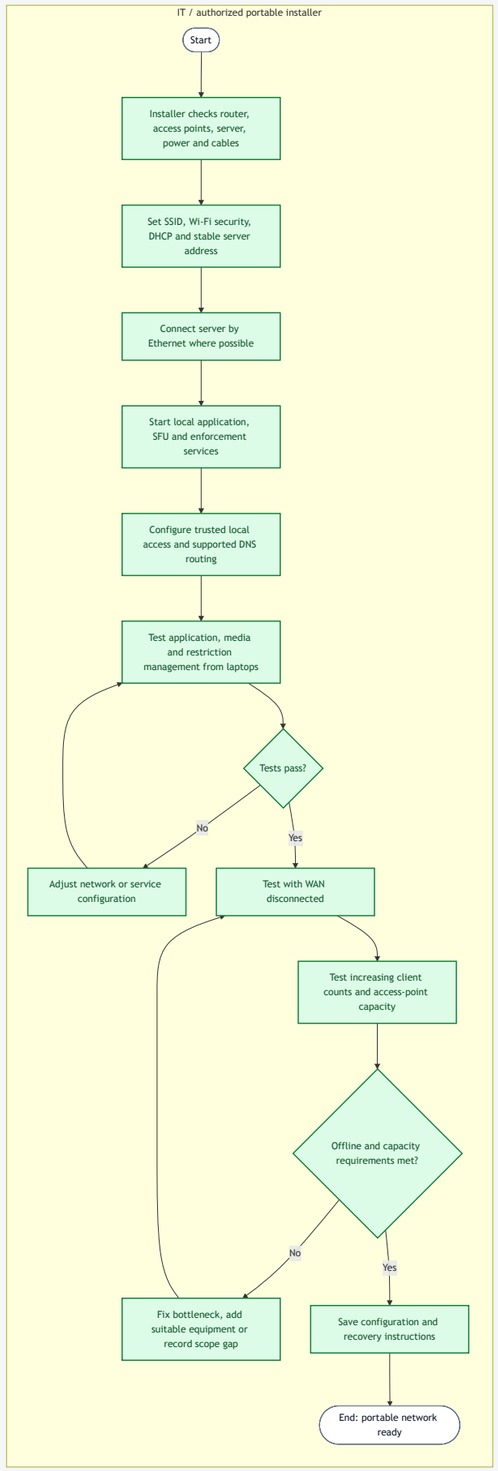
Flow 03 — Prepare portable deployment

Responsibility lanes: user/operator actions and automatic application/media/network actions are grouped separately. Authorization checks are server-side; delegated class automation uses recorded scope approval.

Workflow classification: MUST* — proposed MoSCoW label for source deployment obligations.

Covers OR01–OR12. A teacher laptop may host the services if suitable; a separate local mini-PC/server is another option. Browser-only students do not mean server-free deployment.

Diagram 23



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

The example IP addresses in the source are examples, not fixed requirements. One router is not automatically sufficient for 150 active viewers; capacity and access-point needs must be measured.

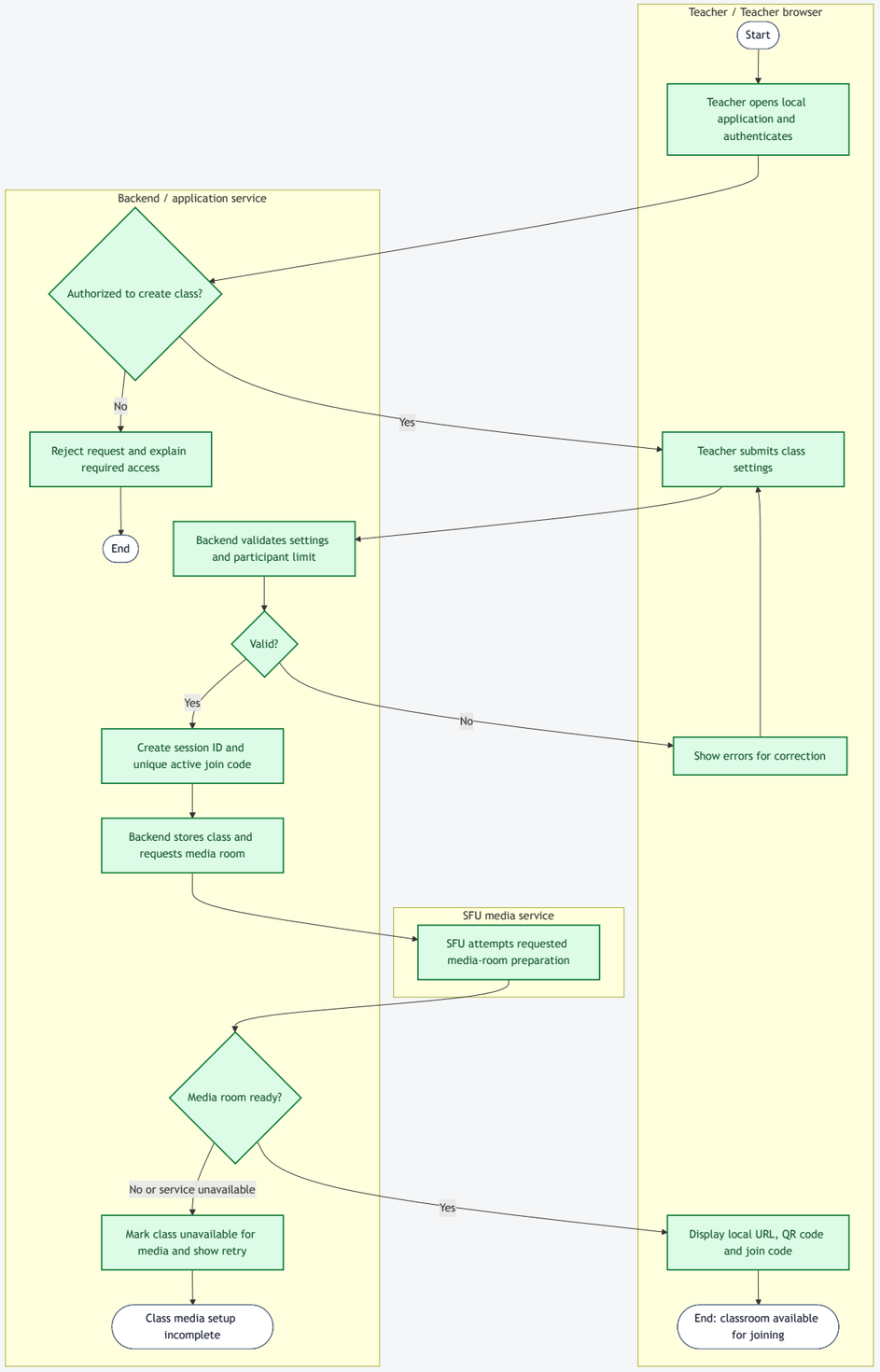
Flow 04 — Teacher creates a classroom

Responsibility lanes: user/operator actions and automatic application/media/network actions are grouped separately. Authorization checks are server-side; delegated class automation uses recorded scope approval.

Workflow classification: MUST (process steps support required capabilities).

Covers T01, T08 and classroom administration. Authentication is needed to enforce teacher authority; the source does not prescribe a particular login provider or token format.

Diagram 24



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

URL locates the application; **QR code** is a convenient way to open joining information; **join code** identifies/admit access to the session. A QR code does not create network connectivity by itself.

Flow 05 — Student joins and is admitted

Responsibility lanes: user/operator actions and automatic application/media/network actions are grouped separately. Authorization checks are server-side; delegated class automation uses recorded scope approval.

Workflow classification: MUST (process steps support required capabilities).

Covers S01–S02, S07, T05–T06; optional S08. Mandatory manual approval for every student is a design choice. The system must support teacher approval/rejection, but the default admission mode needs agreement.

Required join code and proposed admission modes

S02 requires entry of a classroom join code in both modes below. Teacher approval is an additional admission gate, not a replacement for code validation. Supporting both selectable modes is a proposal under D02; the required teacher approve/reject capability remains in scope. There is no approval-only, code-free joining path in this draft. A future QR-prefilled-code convenience would need explicit agreement about how S02 is satisfied; the current flow asks the student to enter the code.

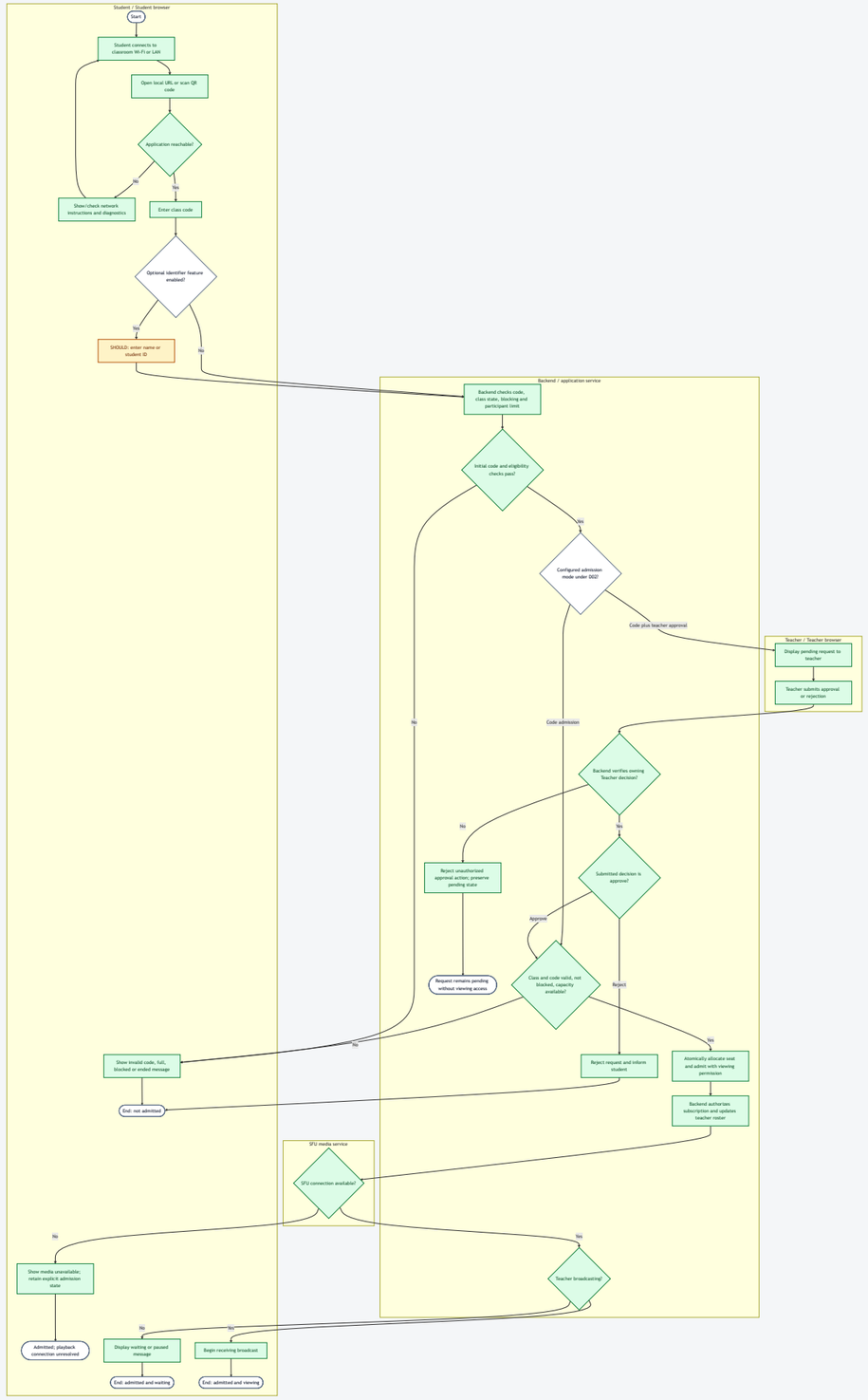
Proposed mode	Process	When viewing access is granted
Code admission	Student enters code → backend validates active class/code, block state and capacity → backend admits	Only after current checks pass; a valid code alone cannot bypass a full/closed class or a block
Code + teacher approval	Student enters code → backend checks eligibility → request waits for owning Teacher → Teacher approves/rejects → backend rechecks eligibility and capacity	Only after valid owning-Teacher approval and final backend checks; pending/rejected requests receive no subscriber grant

Final checks must be atomic with admission/capacity allocation to avoid over-admitting simultaneous requests. If a code expires/rotates, the class ends, a block is added or capacity fills while approval is pending, do not admit based on the old check. Define pending-request timeout/cancellation and reserved-seat semantics in D02; proposed default is no seat reservation before admission. A rejected unauthorized approval attempt must not mutate the pending request as though the owning Teacher rejected it; deny the action and retain its actual admission state.

Admission test	Proposed pass/fail check	References
TC-ADM01	Code admission: valid active code plus available capacity admits; invalid/expired/closed code is rejected	S02, TC-S02, F05
TC-ADM02	Code + approval: valid code creates pending request with no viewing grant; owning Teacher approval admits only after final checks	T06, TC-T06, F05
TC-ADM03	Owning Teacher rejection prevents viewing; pending/cancelled/timed-out requests do not obtain media access	T06, TC-T06; timeout policy pending D02
TC-ADM04	Student, unrelated Teacher, or unscoped administrator cannot approve a request; denied attempt does not change its pending state	NAC16, TC-AUTH, F05/F08
TC-ADM05	While approval waits, end class / expire code / add block / fill capacity in separate trials; subsequent approval does not bypass final eligibility checks	T06, S02, TC-AUTH
TC-ADM06	Two simultaneous requests compete for the final seat; at most one receives admission/media access under the agreed capacity rule	Classroom limits, S02, TC-S02
TC-ADM07	A missing join code is rejected in either mode; selecting approval mode does not create a code-free endpoint	S02, TC-S02

These are test specifications, not executed tests. Criteria approval remains Pending confirmation and results remain Not run. Link implementation of these checks to existing CC-003, CC-006 and CC-007 and their relevant verification items; no completed feature is inferred.

Diagram 25



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

The diagram assumes the teacher makes a decision on a pending request; cancellation/timeouts require a final design. Names and class codes are not strong identity proof. Blocking persistence must be defined, particularly if students join anonymously.

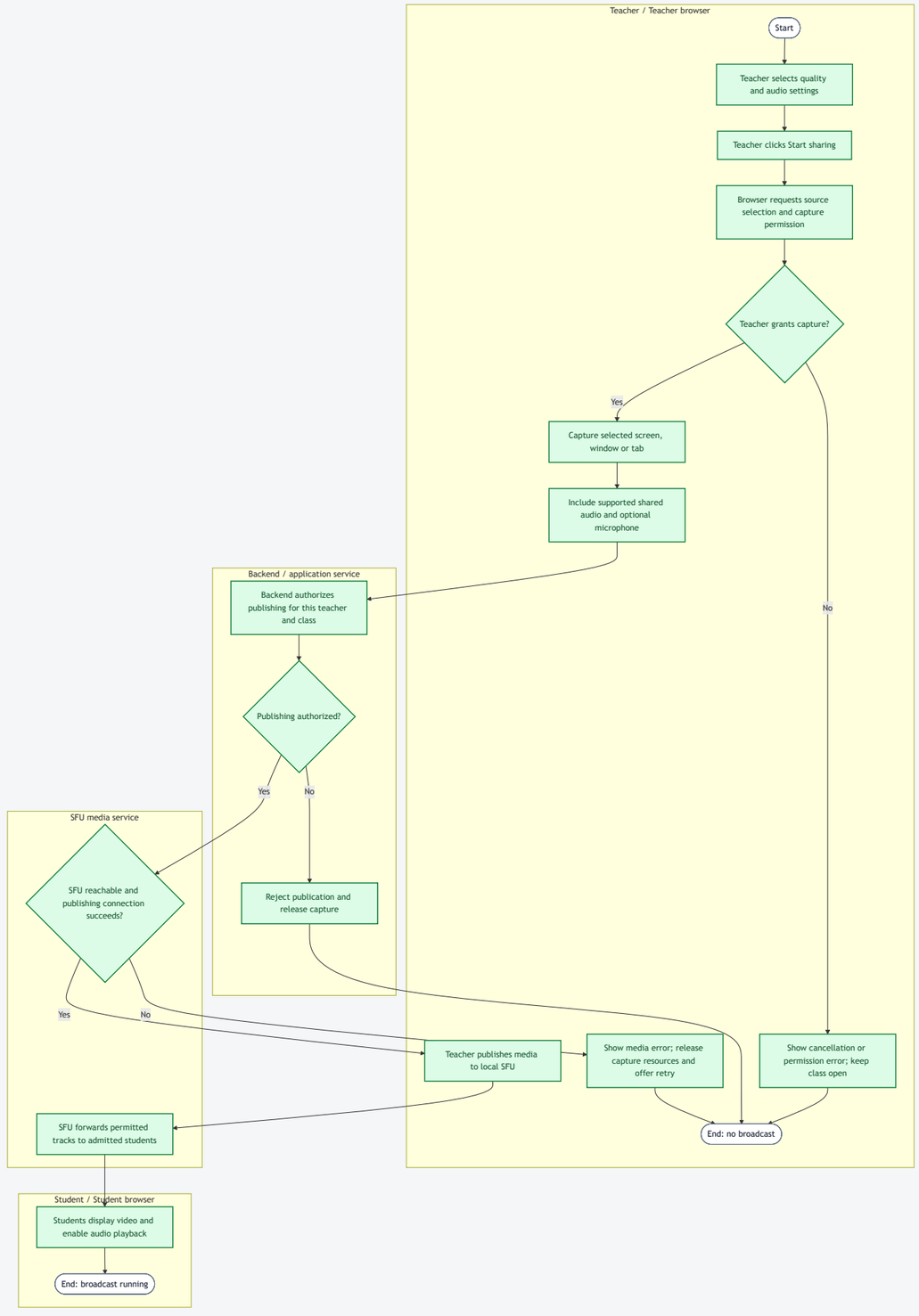
Flow 06 — Start screen and audio broadcasting

Responsibility lanes: user/operator actions and automatic application/media/network actions are grouped separately. Authorization checks are server-side; delegated class automation uses recorded scope approval.

Workflow classification: MUST (process steps support required capabilities).

Covers T02–T04, T07, S03–S04. WebRTC carries the real-time media. Signaling establishes the connection. The SFU receives teacher media and forwards it to students; the database stores application records, not the live video stream.

Diagram 26



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

Capture requires the browser-supported secure context and user permission specified in the source. Audio support depends on operating system, browser and source selection. If audio is unavailable, show that limitation rather than claiming it is being shared. Agree which combinations satisfy the requirement.

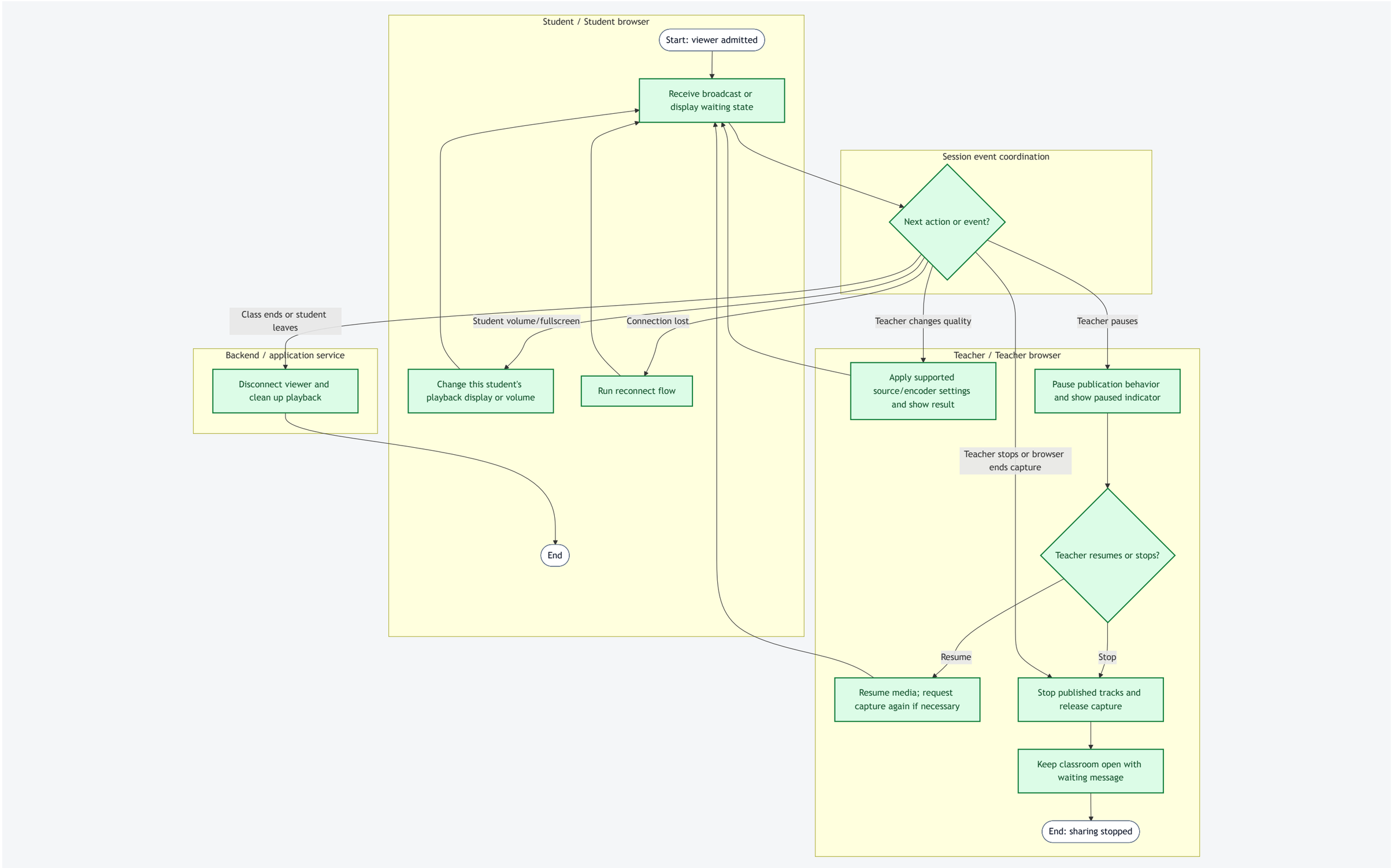
Flow 07 — View, pause, resume and stop sharing

Responsibility lanes: user/operator actions and automatic application/media/network actions are grouped separately. Authorization checks are server-side; delegated class automation uses recorded scope approval.

Workflow classification: MUST (process steps support required capabilities).

Covers T02, T07, S03–S05, S07. A student changes their own playback controls. A teacher changes the published stream. Stopping sharing is different from ending the classroom.

Diagram 27



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

Unresolved: whether pause freezes the last frame or shows a placeholder, whether audio pauses, and whether changing resolution requires new capture. Proposed behavior: clear paused indicator and no continuing live audio. The stop branch does not automatically restore the class policy unless the class is ended or restrictions are separately disabled.

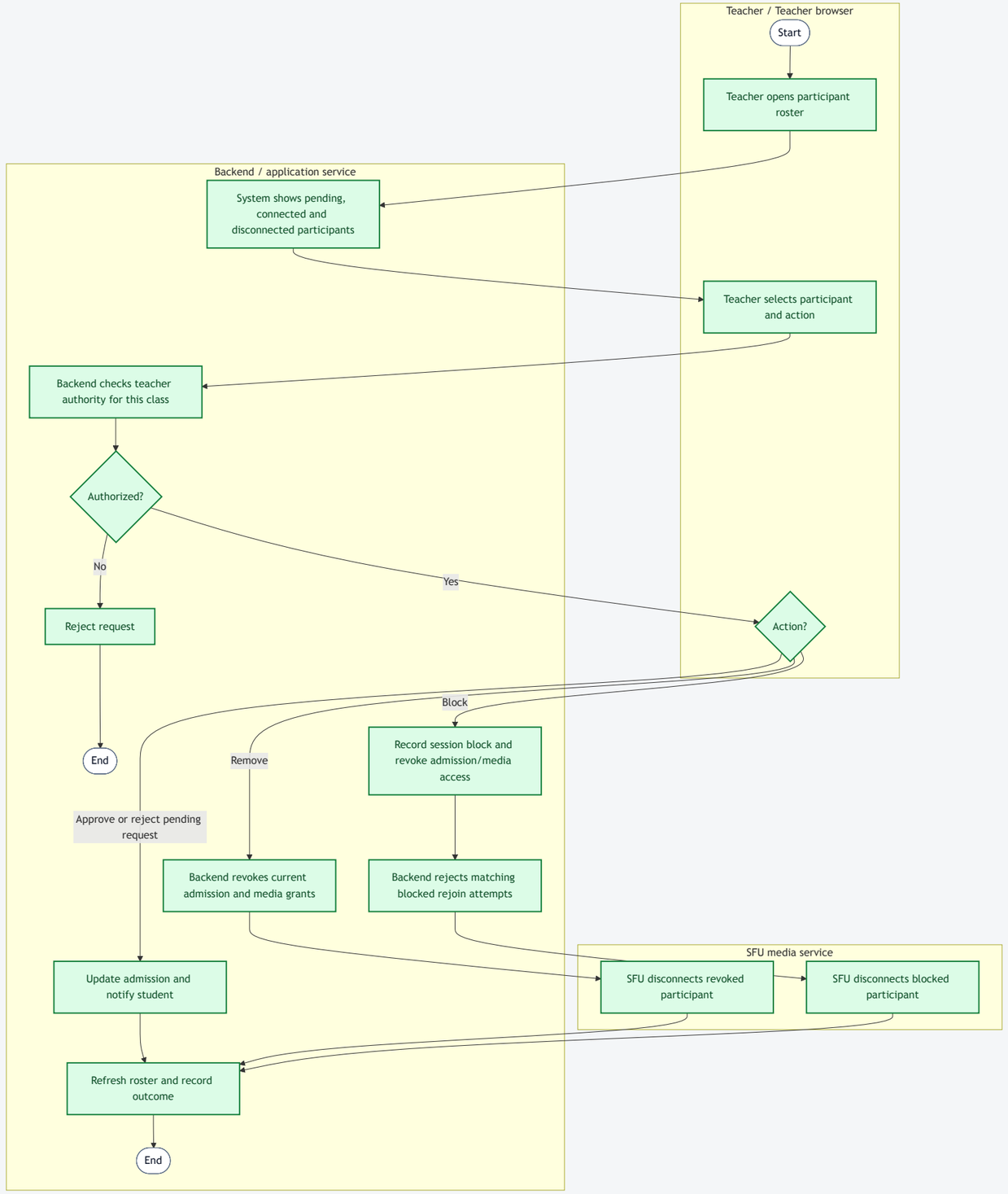
Flow 08 — Teacher monitors, removes or blocks a participant

Responsibility lanes: user/operator actions and automatic application/media/network actions are grouped separately. Authorization checks are server-side; delegated class automation uses recorded scope approval.

Workflow classification: MUST (process steps support required capabilities).

Covers T05–T06 and classroom access control. Monitoring here means connected participants and connection status. Viewing student screens, remote control and monitoring all their activity are not specified.

Diagram 28



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

Proposed distinction: removal ends current participation; blocking also prevents rejoin for the agreed identity and duration. The source requires both actions, but identity and block lifetime need clarification. A browser-only anonymous application cannot guarantee recognition after all browser identity is cleared.

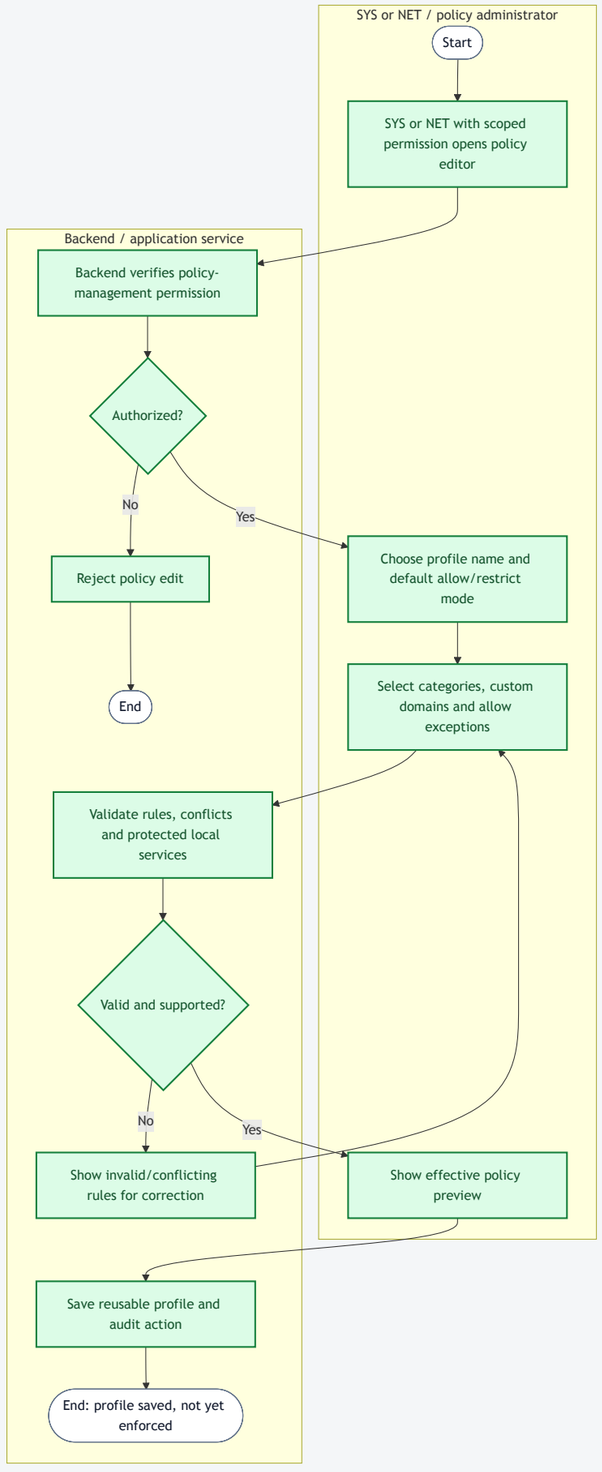
Flow 09 — Administrator creates a policy profile

Responsibility lanes: user/operator actions and automatic application/media/network actions are grouped separately. Authorization checks are server-side; delegated class automation uses recorded scope approval.

Workflow classification: MUST (process steps support required capabilities).

Covers NAC02–NAC04, NAC07–NAC08, NAC16–NAC18. A profile is a reusable configuration, such as Normal Class, Programming Lab or Exam Mode. Saving one does not apply it to the network.

Diagram 29



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

Allow-by-default: public destinations are permitted unless a rule restricts them. **Restricted-by-default:** only explicitly approved destinations are permitted within the adapter's supported scope. Specify exception precedence and domain/subdomain matching before implementation. Neither mode alone proves protection against VPNs or alternate DNS.

Categories are maintained domain collections, not an automatic guarantee of identifying every AI tool. The category checkbox preview in the pasted requirements is an illustration, not an active policy.

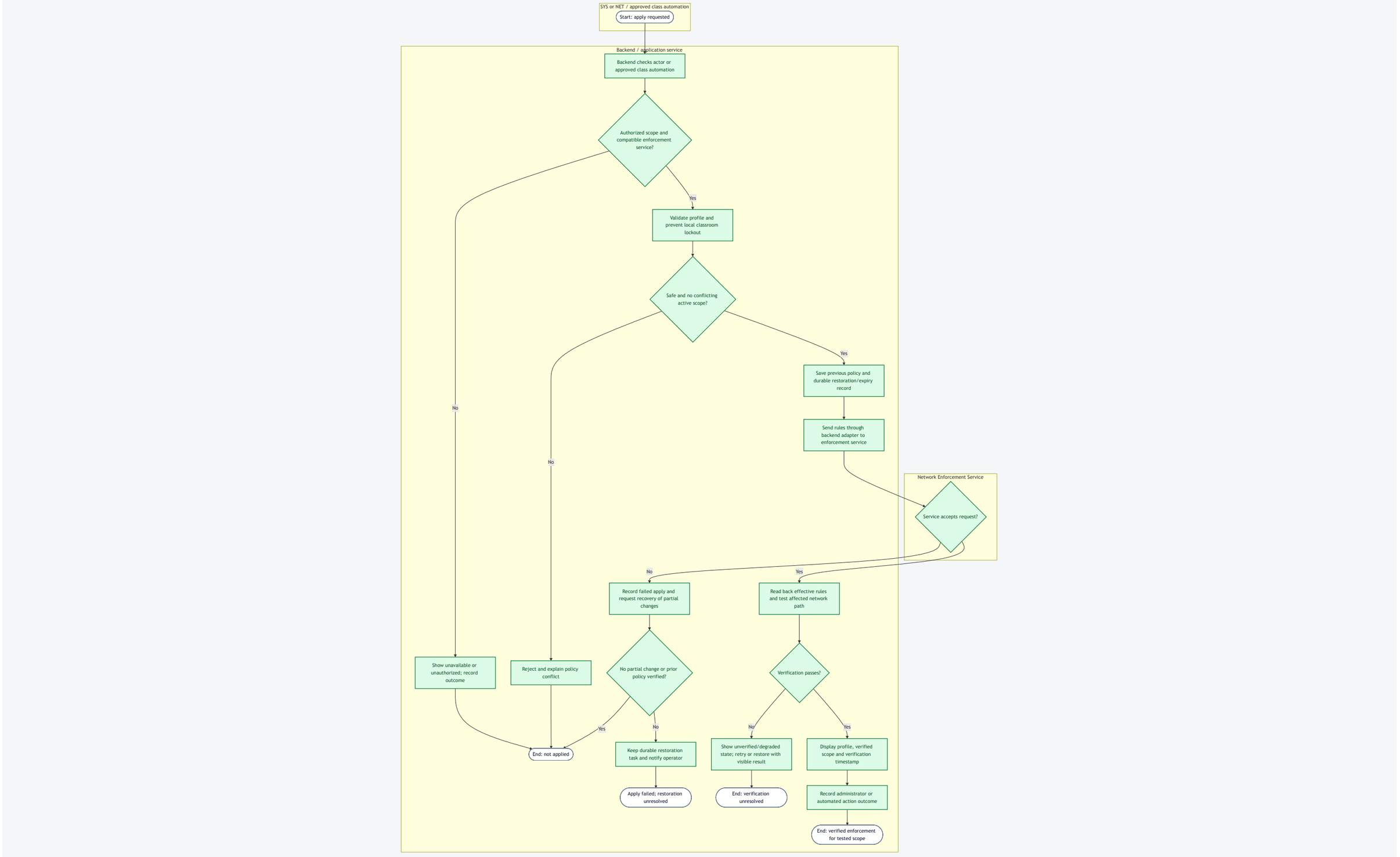
Flow 10 — Apply and verify restrictions

Responsibility lanes: user/operator actions and automatic application/media/network actions are grouped separately. Authorization checks are server-side; delegated class automation uses recorded scope approval.

Workflow classification: MUST (process steps support required capabilities).

Covers NAC01, NAC05, NAC10–NAC11, NAC16–NAC19; CN10/OR12. Trigger: a class starts with an administrator-approved profile, or an authorized administrator requests application. The teacher's start event can trigger preapproved policy behavior without giving the teacher unrestricted policy-edit privileges.

Diagram 30



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

Saving a baseline durably and refusing conflicting scope operations are proposed implementation safeguards. The requirement is reliable apply/status/restore behavior; the storage mechanism is a design choice.

Three different results: profile saved; service accepted changes; restrictions verified on the tested path. Display them accurately. A backend DNS test alone does not establish that every student is using the configured resolver.

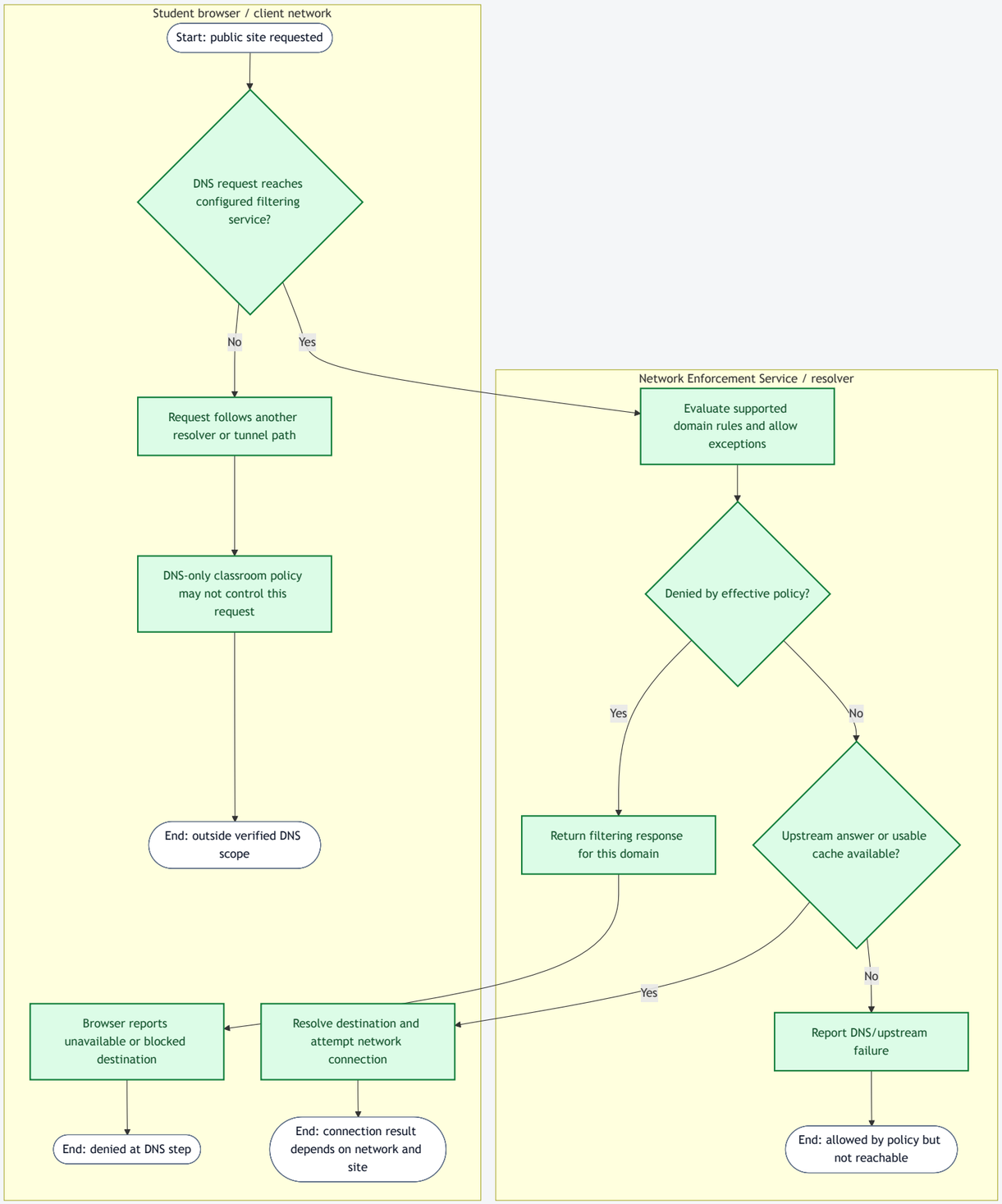
Flow 11 — A student requests a public website

Responsibility lanes: user/operator actions and automatic application/media/network actions are grouped separately. Authorization checks are server-side; delegated class automation uses recorded scope approval.

Workflow classification: MUST (process steps support required capabilities).

This explains what the enforcement component does. It is separate from the student watching the local broadcast. It also explains the supplied DNS-filtering limitations.

Diagram 31



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

This diagram is scoped to DNS lookups. Cached addresses, existing connections, direct IP access and offline agents need separate treatment; policy changes may not instantly terminate existing website connections. A gateway can enforce additional restrictions only where supported and authorized. Do not promise a custom block page for every HTTPS website.

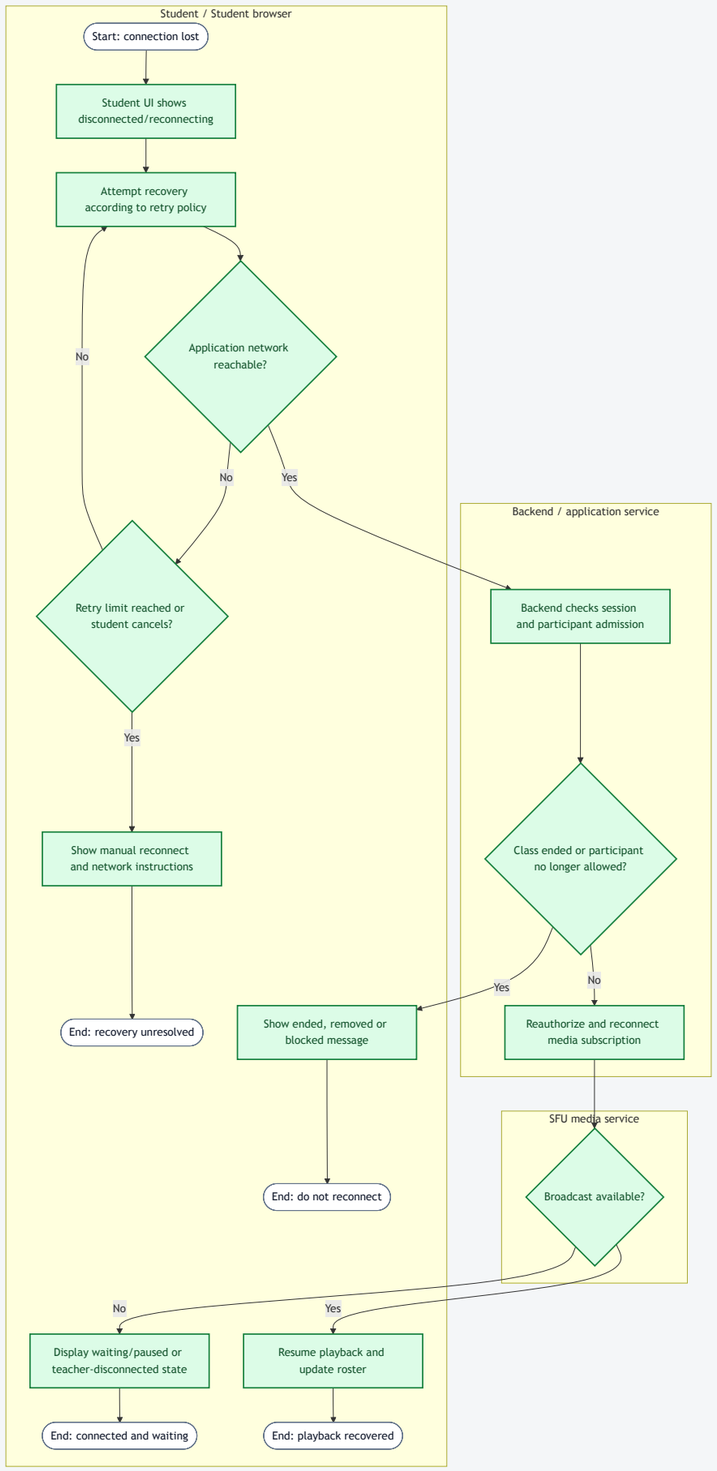
Flow 12 — Connection loss and reconnection

Responsibility lanes: user/operator actions and automatic application/media/network actions are grouped separately. Authorization checks are server-side; delegated class automation uses recorded scope approval.

Workflow classification: MUST (process steps support required capabilities).

Covers S06–S07 and teacher connection status. The source requires reconnection but does not define timeout/retry limits or teacher-offline behavior.

Diagram 32



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

A temporary teacher connection failure should not silently mean that network restrictions are restored or left forever. Proposed behavior: keep explicit class state, use visible recovery status, and retain an independent policy expiry mechanism. Agree the teacher-disconnect grace period before implementation.

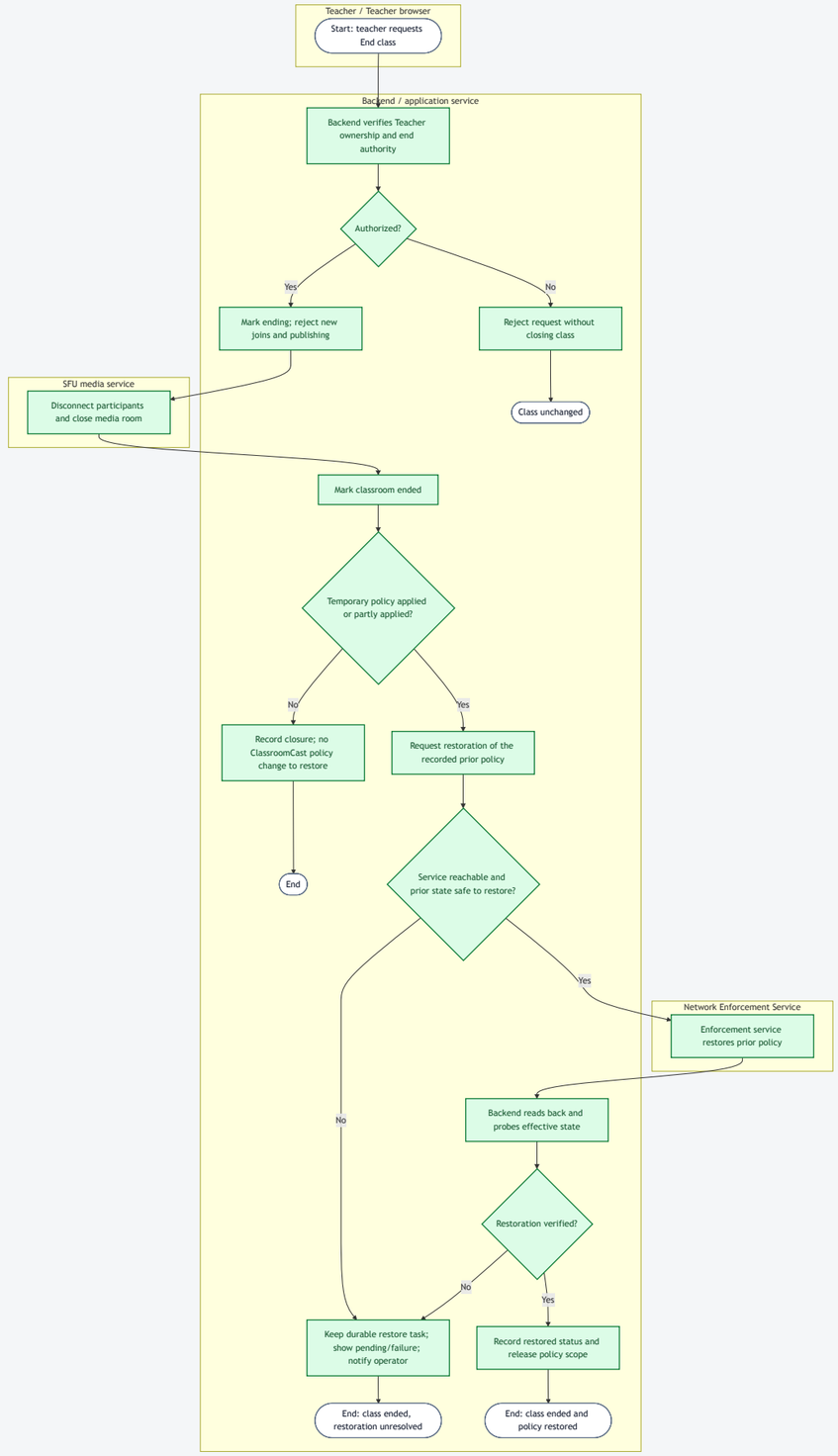
Flow 13 — End a classroom and restore policy

Responsibility lanes: user/operator actions and automatic application/media/network actions are grouped separately. Authorization checks are server-side; delegated class automation uses recorded scope approval.

Workflow classification: MUST (process steps support required capabilities).

Covers NAC06, NAC10–NAC11 and classroom end/disconnect requirements. Closing student access and restoring network policy are separate outcomes; a failed restore must remain visible even after the class has ended.

Diagram 33



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

Proposed safeguard: detect unrelated configuration changes before overwriting a saved snapshot. A simple unrestricted “allow everything” operation is not necessarily correct restoration. Preserve protections that existed before ClassroomCast started the class.

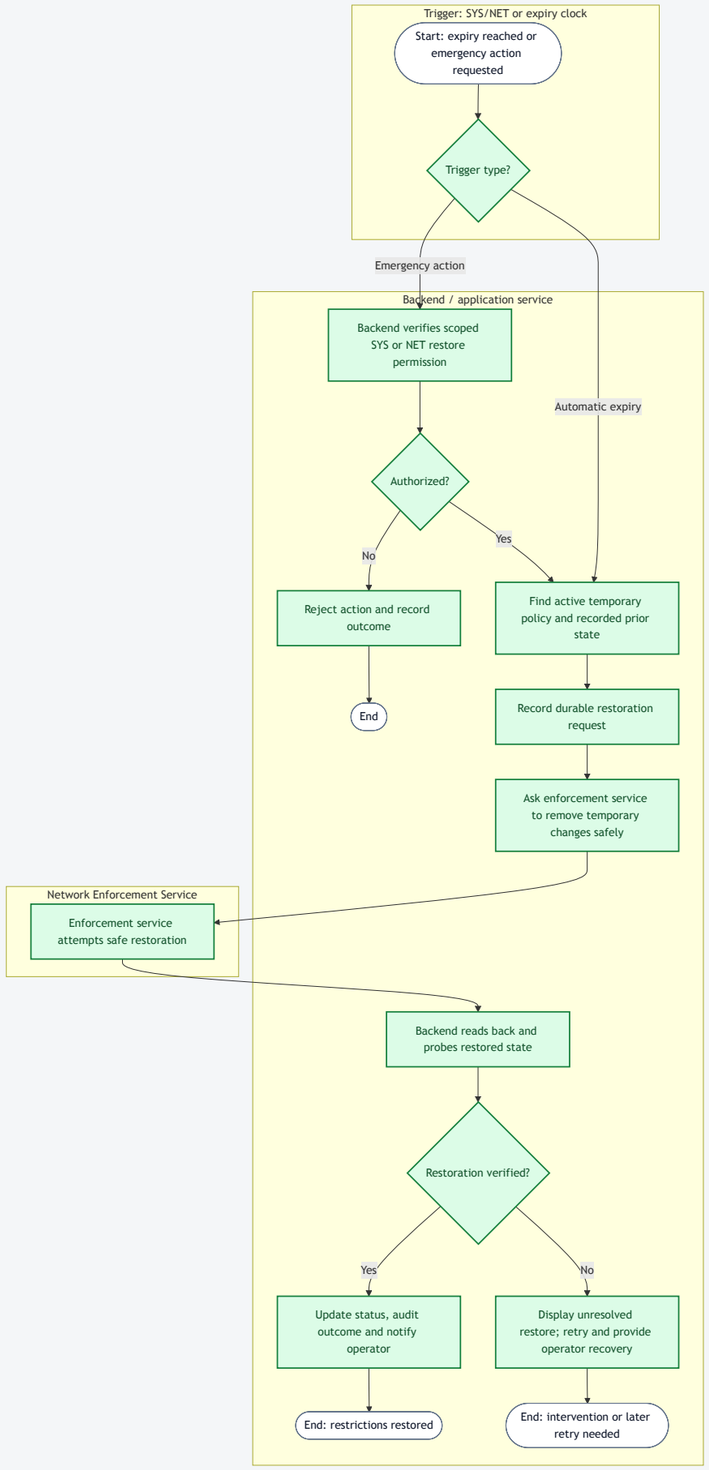
Flow 14 — Temporary expiry and emergency disable

Responsibility lanes: user/operator actions and automatic application/media/network actions are grouped separately. Authorization checks are server-side; delegated class automation uses recorded scope approval.

Workflow classification: MUST (process steps support required capabilities).

Covers NAC12–NAC13 and recovery requirements. Expiry should operate without a teacher browser remaining open. Proposed durable expiry records and a recovery worker support that requirement.

Diagram 34



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

Restoration cannot be guaranteed while the enforcement service is unreachable. Independent service-side expiry may strengthen recovery where supported. Emergency disable should restore ClassroomCast's changes without disabling unrelated campus controls. Whether ending restrictions also ends the classroom is unspecified; proposed behavior is to keep these separate unless the operator ends the class.

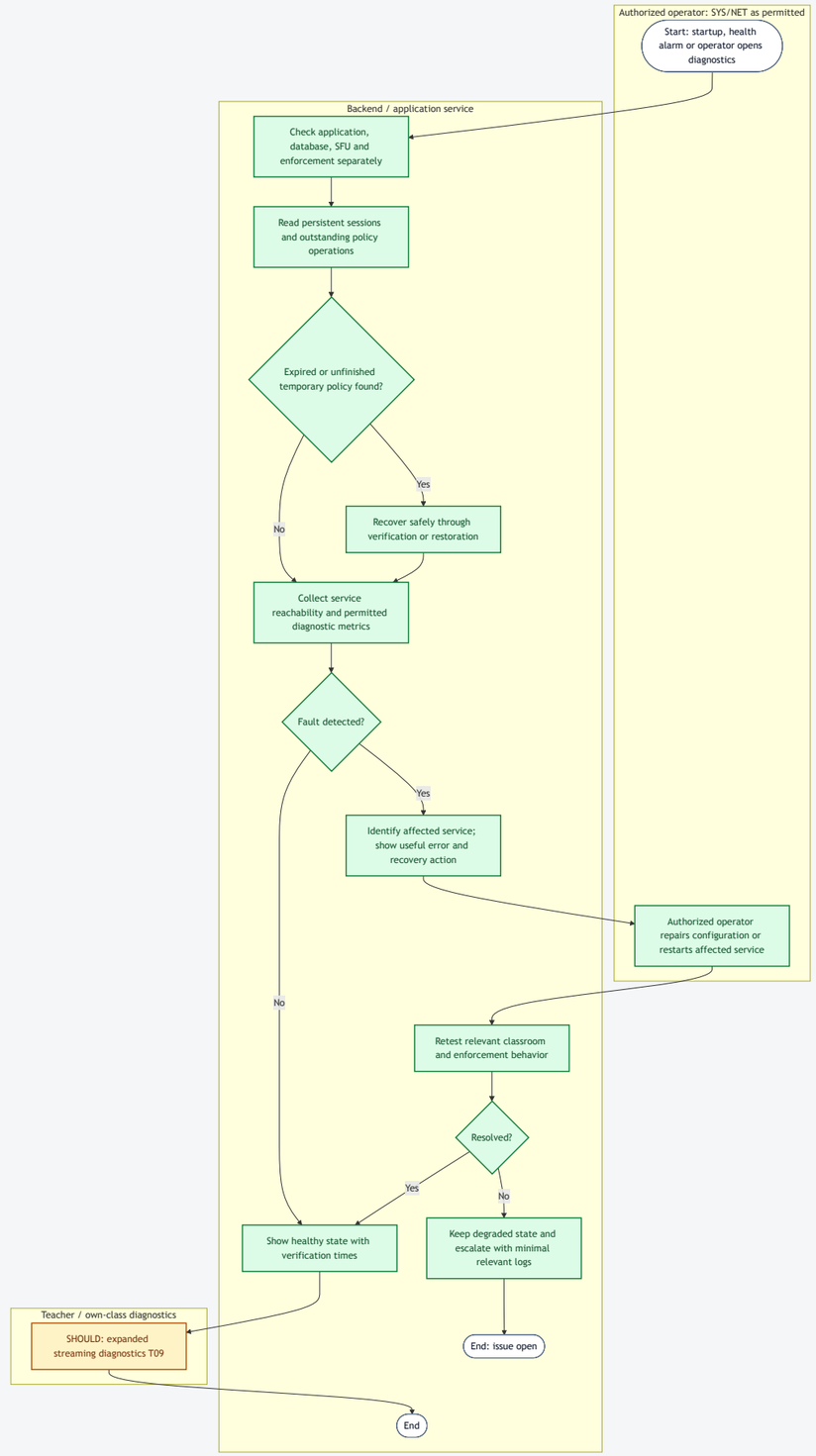
Flow 15 — Health, restart and diagnosis

Responsibility lanes: user/operator actions and automatic application/media/network actions are grouped separately. Authorization checks are server-side; delegated class automation uses recorded scope approval.

Workflow classification: MUST (process steps support required capabilities).

Covers local diagnostic page, health checks, logs and recovery; T09 is Should for expanded streaming diagnostics. Basic health/recovery obligations already exist in the architectural requirements.

Diagram 35



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

Avoid displaying global “Healthy” when only the web page is reachable. Media can fail while the API works; DNS filtering can fail while media works. Logs must avoid secrets and unnecessary browsing-history collection (NAC17/NAC19).

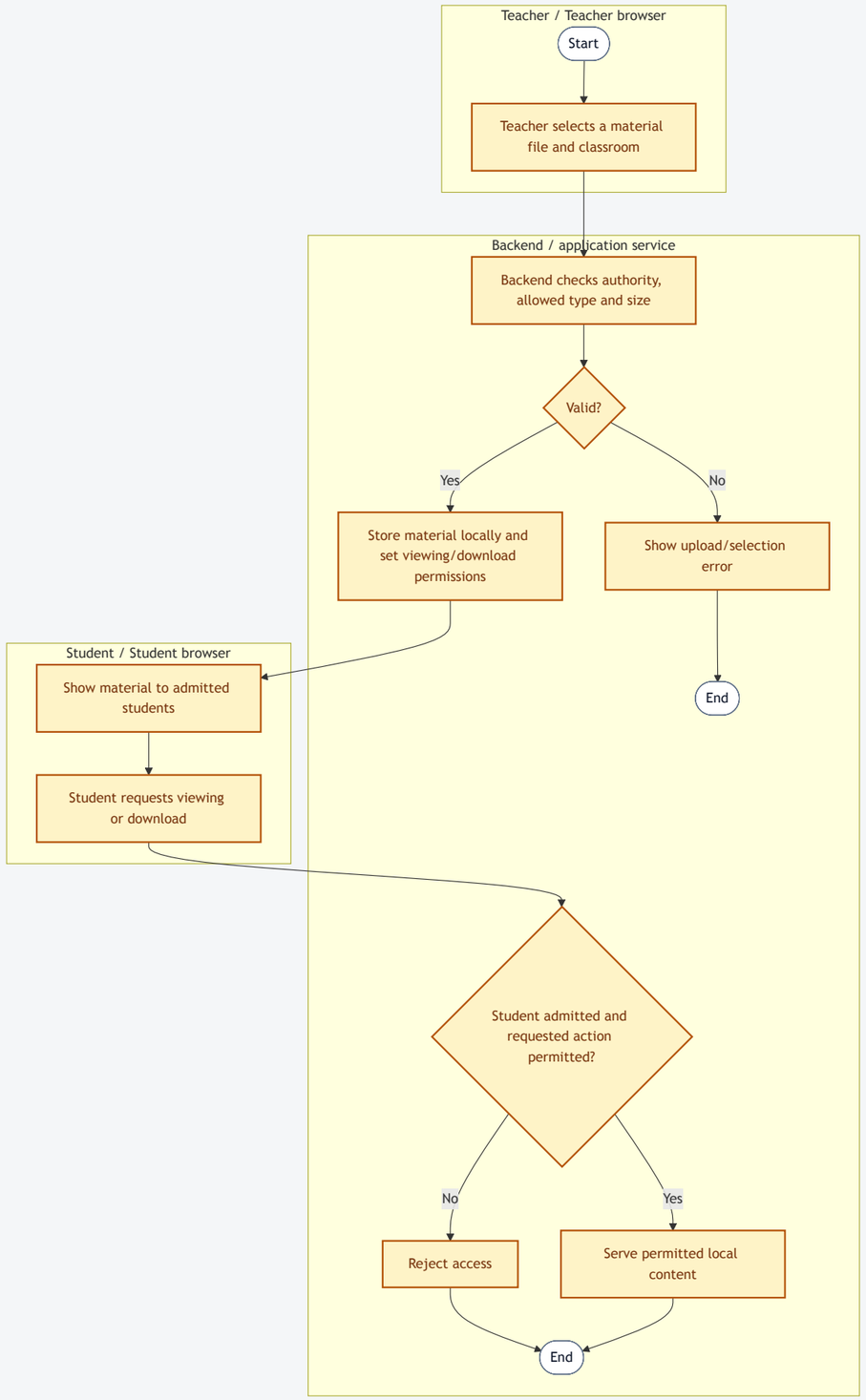
Flow 16 — Share and download materials — Should

Responsibility lanes: user/operator actions and automatic application/media/network actions are grouped separately. Authorization checks are server-side; delegated class automation uses recorded scope approval.

Workflow classification: SHOULD (authorization and validation still required when implemented).

Covers T10 and S09. Displaying a PDF inside a shared screen is already possible through screen capture. Uploading a file and giving students download permission is an additional application workflow. The exact upload/transfer model is a proposed interpretation to confirm.

Diagram 36



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

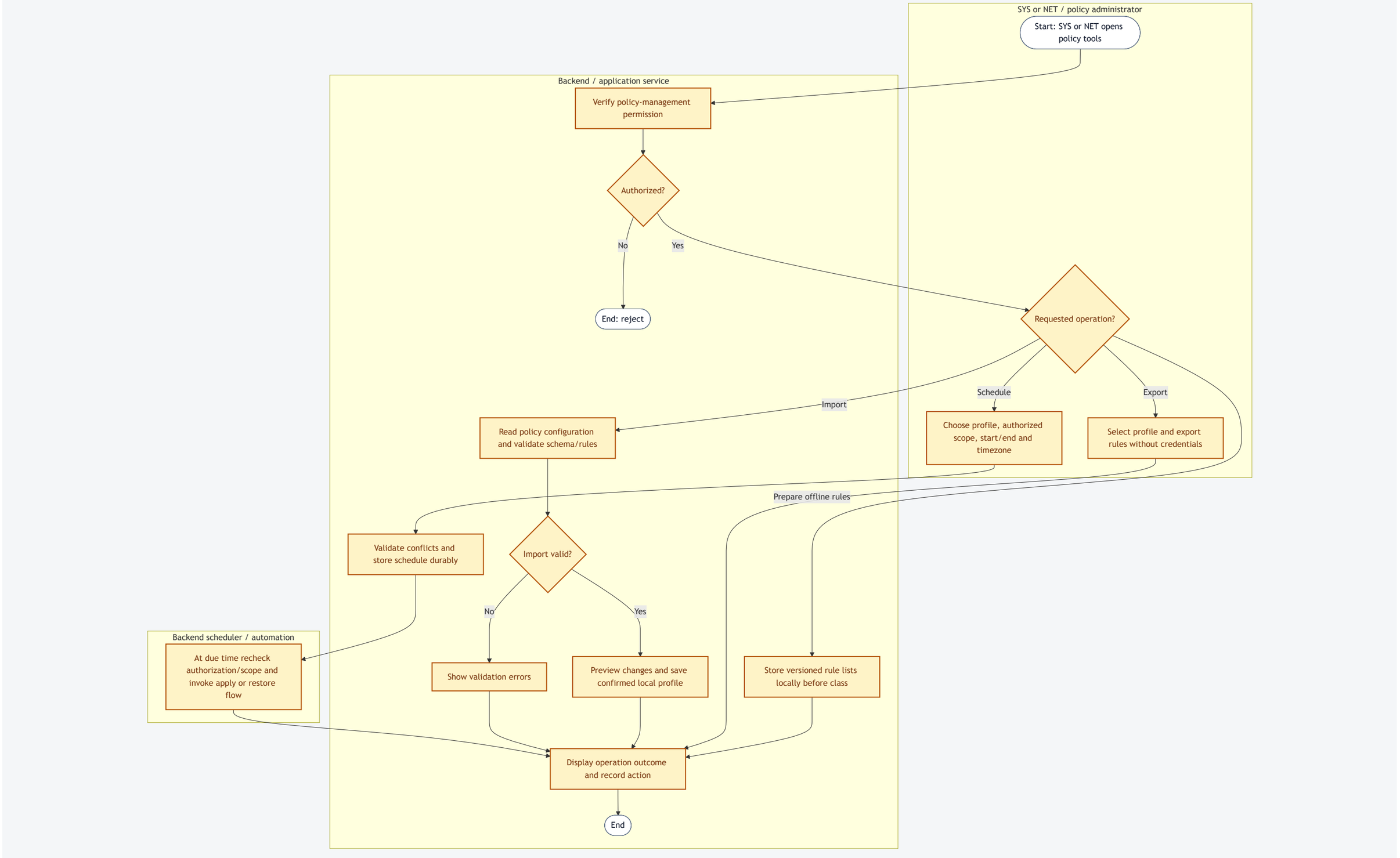
Flow 17 — Policy schedules and configuration transfer — Should

Responsibility lanes: user/operator actions and automatic application/media/network actions are grouped separately. Authorization checks are server-side; delegated class automation uses recorded scope approval.

Workflow classification: SHOULD (authorization and validation still required when implemented).

Covers NAC09, NAC14, NAC20. Local rules are also necessary for the broader offline operation requirement; clarify that overlap rather than allowing the global requirement to disappear behind a Should label.

Diagram 37



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

Importing or scheduling a profile does not itself prove enforcement. Scheduled operations must still verify outcomes. Timezone and clock behavior during offline operation need agreement.

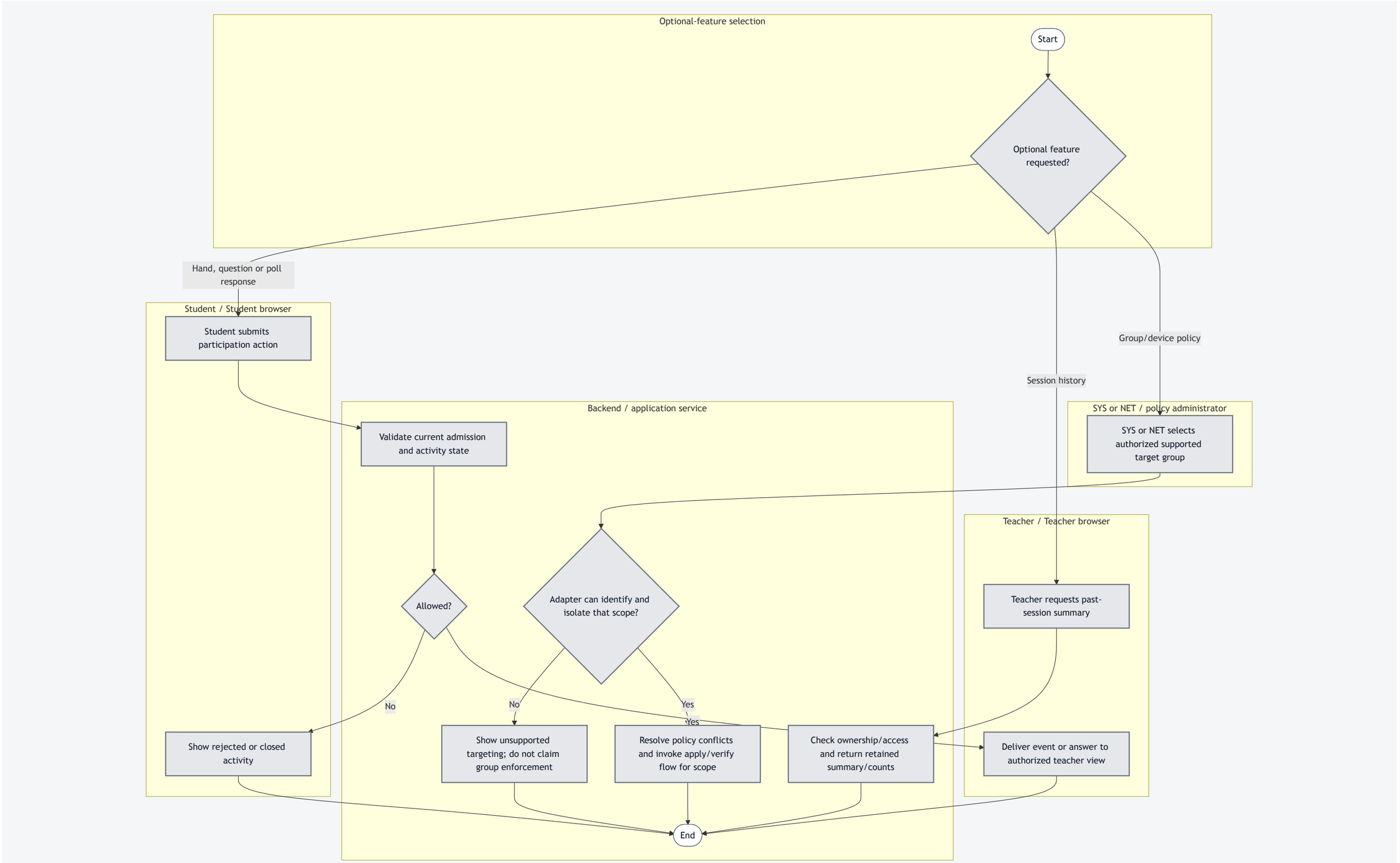
Flow 18 — Optional participation, history and group policies — Could

Responsibility lanes: user/operator actions and automatic application/media/network actions are grouped separately. Authorization checks are server-side; delegated class automation uses recorded scope approval.

Workflow classification: COULD (authorization and scope checks still required when implemented).

Covers S10, T11, NAC15. These are three distinct optional features, shown as separate branches so they do not become hidden Must work.

Diagram 38



Rendered from the included Mermaid source. Draft design; not evidence of implementation or approval.

For polls, question creation/open/close/results behavior needs refinement; the source lists the feature but does not specify the complete poll product. Session history does not imply recording the teacher's screen. Per-device network identity cannot automatically be inferred from a browser nickname, especially behind shared network translation.